

# UGED1111 Logic

## Basic Concepts I

CHAN, Hei Man

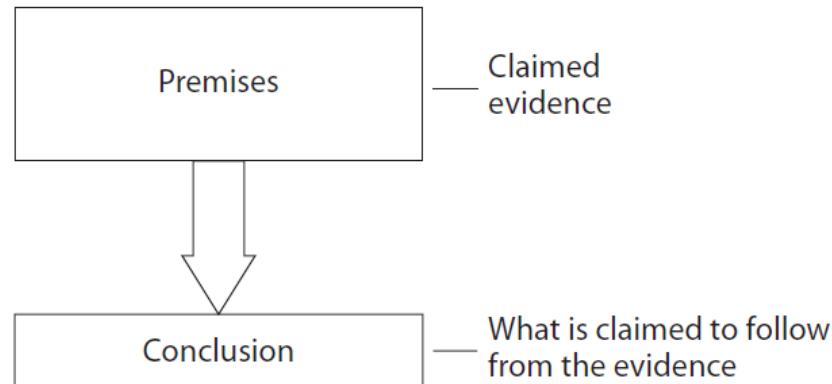
# Agenda



- **Basic elements and structure of an argument**
- **How to recognize an argument**
- **Deductive argument and Inductive argument**

# Basic Form of an Argument

- ◆ The **premises** are the **statement(s)** that set forth the **reasons or evidence**.
- ◆ The **conclusion** is the **statement** that the evidence is claimed to support/ justify.



# Statement

- Premise(s) and Conclusion must be **statements**
- **Declarative sentence** that is ***either true or false***
- Grammatical

Example:

1. Today is 17<sup>th</sup> May 2024.
2. 韓國瑜是韓國人
3. Water is H<sub>2</sub>O.



# Discussion 1

## Are they statements?

1. How is today's weather?
2. Snow white slow Is
3. 如果全球氣溫再上升五度，香港就會消失。
4. 球波一係入、一係唔入。
5. Leave me alone!
6.  $2+2=2$

# Fact check?

與您同賀美心60周年·免費送紀念版月餅

**不要再傳了！  
是2016年的！**

只要香港身份證號碼有  
"60"或"18"兩組相連數字，  
抑或身份證姓名有  
"美"或"心"字

到美心月餅 Pop-up Store  
免費獲贈「美心60周年紀念  
版月餅」1盒(價值\$248)，  
每日限量100盒！

8月13日-9月4日期間的星期六、日



求驗傳媒

· 8月8日 ·

年份的重要性

美心又送月餅？這幾天Whatsapp流傳的圖，大家傳得不亦樂乎，但大家可會覺得該圖有點似曾相識呢？

其實，那圖美心是2016年的「60周年」優惠，而美心不是譚詠麟，不會年年都「60歲」。在2017年，也有網民把這個2016年的優惠誤傳，結果美心也有澄清。

今年已經是2019年，謠言之可以無限復活，最主要原因是該圖沒有年份，也沒有網站資料供參考。

所以，還是那一句，任何沒有年月日及資料來源的資訊，大家都要小心處理，不要輕易轉發。

2016年的新聞：

<https://www.weekendhk.com/462909/dining/美心月餅-免費-60周年紀念版/>

# Truth

***Truth is the attribute of those statement that assert what really is the case (Copi, pp26)***

***\*Corresponding to the fact?***

**Example:**

- 1. Today is Monday.**
- 2. There is a cat on the Bed.**
- 3. There is no female in this class.**



# Standard Form of an Argument

Premise<sub>1</sub>

Premise<sub>2</sub>

Premise<sub>3</sub>

...

Premise<sub>n</sub>  $(n \geq 1)$

---

Conclusion

# Typical Example



1. Tom is a CUHK Students. (Premise)
  2. All CUHK Students are smart.(Premise)
- 

3. Tom is smart. (Conclusion)

# Premise Indicators

**Typical premise indicators are**

■ Since ; in that; seeing that; as indicated by; may be inferred from; for the reason that; because; as in as much as; for given that; owing to; etc

■ 由於、基於、理由是、因為; etc

**Example:** 由於GPA 高可以得到獎學金，我一定要好好讀書。

**Premise:** GPA 高可以得到獎學金。

**Conclusion:** 我一定要好好讀書。



# Conclusion Indicators

**Typical conclusion indicators are**

■ Therefore; accordingly; entails that; wherefore we may conclude; hence; thus; it must be that ;it follows that; consequently; for this reason; implies that; etc

■ 因此、所以、故此、結論是, etc

**Example:**只有信耶穌才得永生，所以你應該信耶穌。

**Premise:**只有信耶穌才得永生。

**Conclusion:**你應該信耶穌

# Hidden Premise

**Hidden Premise: A premise which assumed in the argument but not stated explicitly**

**Example:**

只有信耶穌才得永生，所以你應該信耶穌。

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# Recap



- **Statement – either true or false**
- **We will not say an argument is “true” or “false”.**
- **\*\*\* “statement” and “proposition”?**

# Question

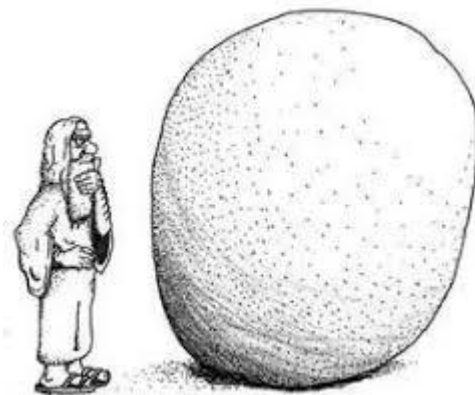
Is this a statement?



# Example

## 石頭論證

如果上帝無法造出自己舉不起來的石頭，那上帝不是全能的。如果上帝可以造出自己舉不起來的石頭，上帝也不是全能的。要不上帝可以造出自己舉不起來的石頭，要不上帝無法造出自己舉不起來的石頭。所以上帝不是全能的。

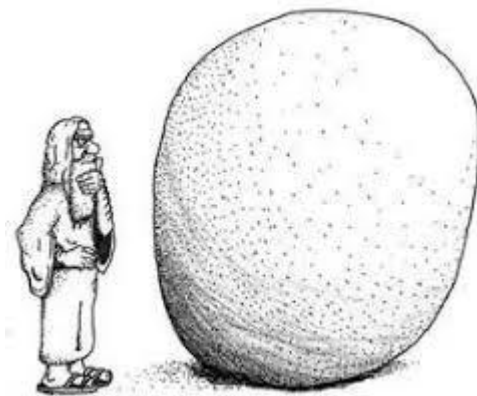


# Example - standard form

## Standard form

1. 如果上帝無法造出自己舉不起來的石頭，上帝不是全能的。
2. 如果上帝可以造出自己舉不起來的石頭，上帝不是全能的。
3. 上帝可以造出自己舉不起來的石頭或上帝無法造出自己舉不起來的石頭。

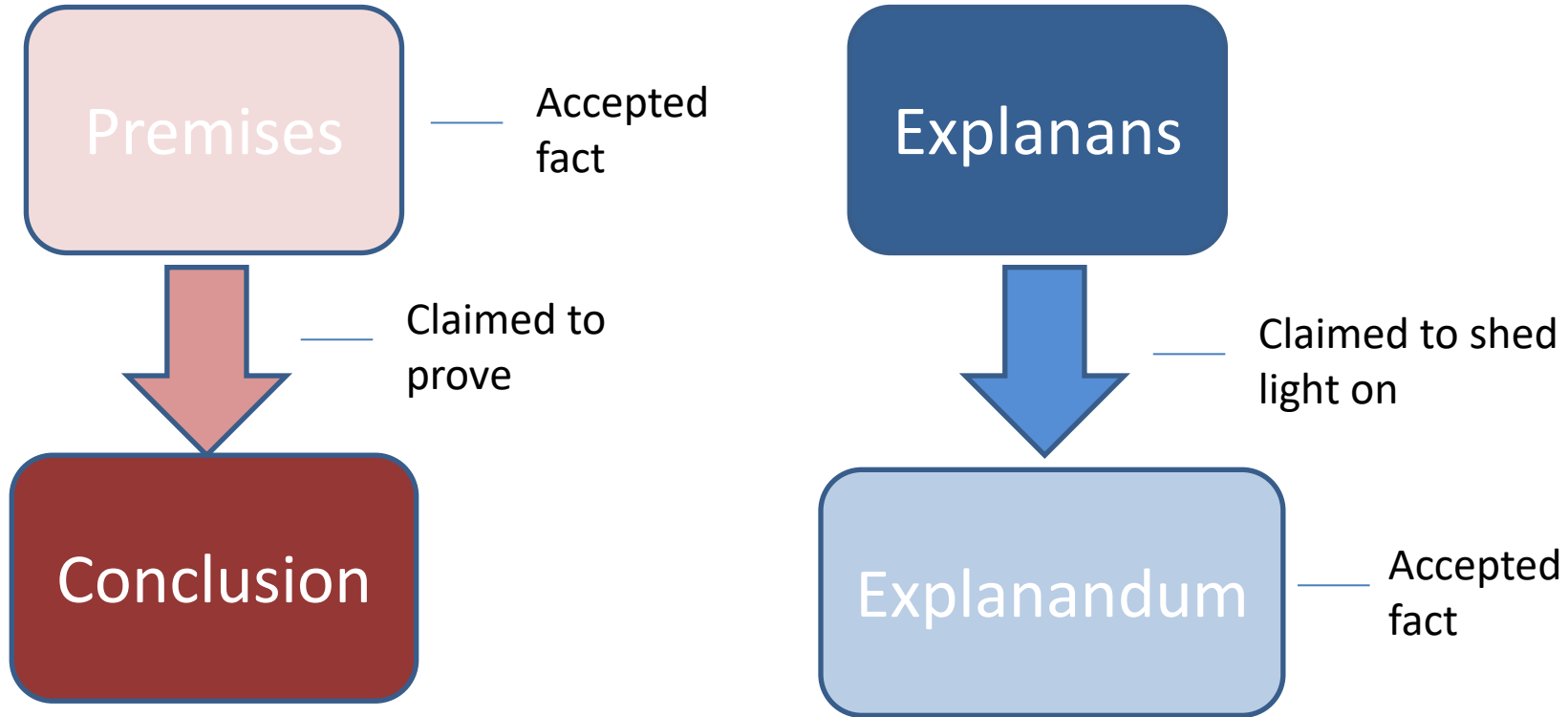
- 
4. 上帝不是全能的。



# NOT -Argument

- ❑ A Single statement
- ❑ Conditional sentence  
(e.g. if it is raining, Tom will bring an umbrella. )
- ❑ A set of statements not intend to support any claim
- ❑ Advice / Warning/ Illustrations, etc
- ❑ Causal explanation

# Argument Vs Explanation



# Explanation



Explanans: statement(s) which is claimed to give the explanation.

Explanandum: statement which describes the accepted fact.



# Argument Vs Explanation

## □ Example 1:

這隻狗有跳蚤，因為牠身上有在蠕動的黑色點，而且你還可以看見牠不停在抓癢。

## □ Example 2

這隻狗有跳蚤，因為昨天帶牠去公園時牠和野狗一起玩，還在草地滾來滾去。

# Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.

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**Basic Concepts II**

# Evaluate an argument

## Standard

- ❑ Whether the premise(s) support the conclusion
- ❑ Whether the premise(s) are true
  
- Different standard of Evaluation for Deductive Argument and Inductive argument
  - Validity and Soundness
  - Strength and Cogency

# Types of Argument

## Deductive argument (演繹論証)

1. 所有中大學生都長得好看。
  2. 小明是中大學生。
- 
3. 小明長得好看。

## Inductive argument (歸納論証)

1. 96 % 的中大學生都長得好看。
  2. 小明是中大學生。
- 
3. 小明長得好看。

# Types of Argument

## ◆ A **deductive argument** (演繹論証)

incorporates the claim that it is **impossible** for the conclusion to be false given that the premises are true.

## ◆ An **inductive argument** (歸納論証)

incorporates the claim that it is **improbable** that the conclusion be false given that the premises are true.

# Deductive Argument

## Example 1:

1. 中年的定義是人均壽命的一半。
  2. 人均壽命百二歲。
- 
3. 六十歲是中年。

## Example 2:

1. 所有貓都有嘴。
  2. Hello Kitty 是貓。
- 
3. Hello Kitty 有嘴。





# Inductive Argument

## Example 1.

1. 十個光頭九個富。
  2. 劉小煒是光頭的。
- 

3. 劉小煒是富翁。

## Example 2

1. 今天亞視存在。
2. 昨天亞視存在。

...

---

3. 亞視會永遠都存在。



# Concepts of Possibility



**What does it mean when we say something is “possible” or “impossible”?**

**“It is impossible for us to travel to Jupiter”**

**當男神/女神跟你說「我們不可能」**

# Concepts of Possibility



- **Technological Possibility**

- ❑ **Not exceed the limitation of current technology**

- **Empirical Possibility**

- ❑ **Not against any laws of nature**

- **Logical Possibility**

- ❑ **Not against fundamental laws of logic**

# Concepts of Possibility

## Technological Possibility

-Not exceed the limitation of  
current technology

Example:

It is impossible for us to  
travel to Jupiter.



# Concepts of Possibility

## Empirical Possibility

--Not against any laws of nature

e.g. 永動機



# Concepts of Possibility

## Logical Possibility

- Not against fundamental laws of logic
- Not involve any contradictions

### Example

- Rounded-square





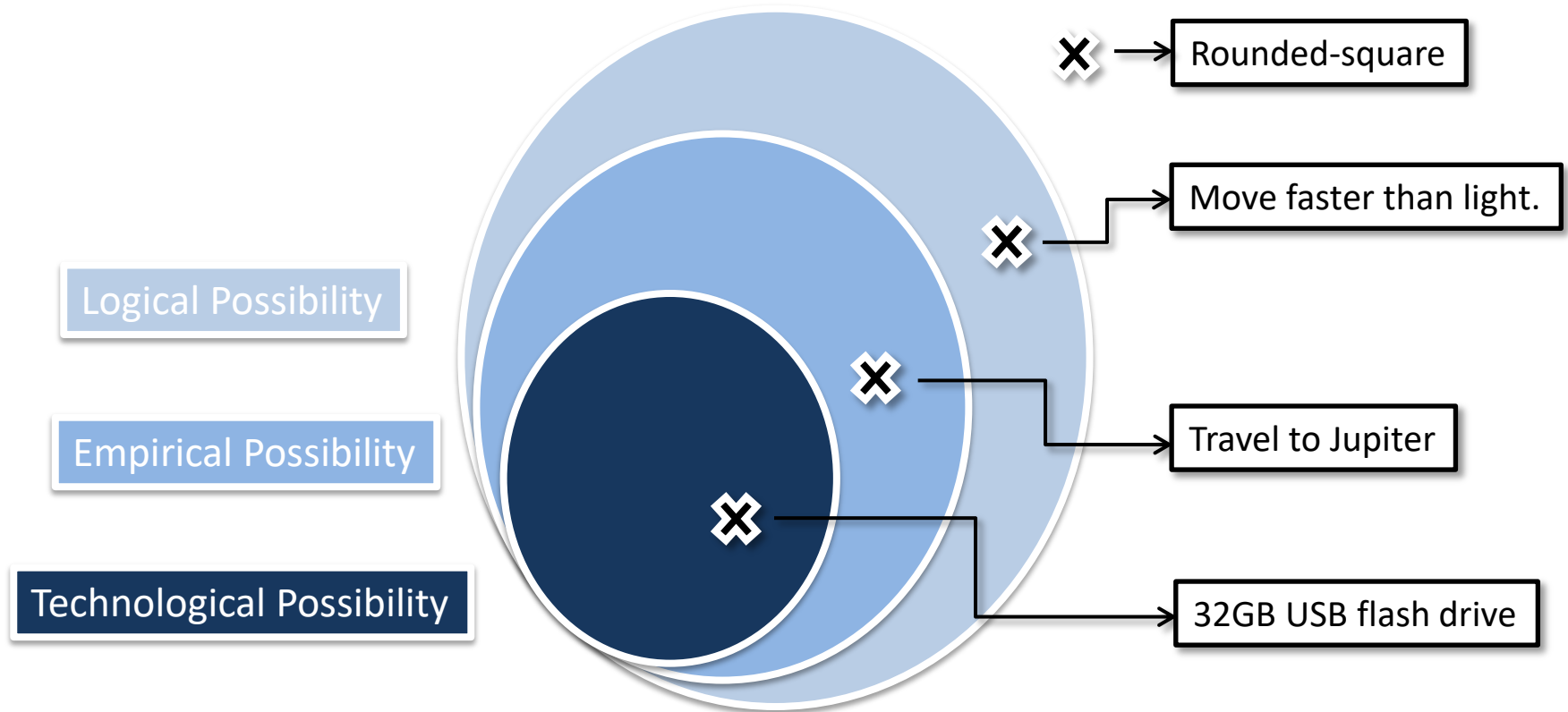
# Concepts of Possibility

1. 如果上帝不可能造出自己舉不起來的石頭，上帝不是全能的。
2. 如果上帝可能造出自己舉不起來的石頭，上帝不是全能的。
3. 上帝可能造出自己舉不起來的石頭或上帝不可能造出自己舉不起來的石頭。

---

上帝不是全能的。

# Concepts of Possibility



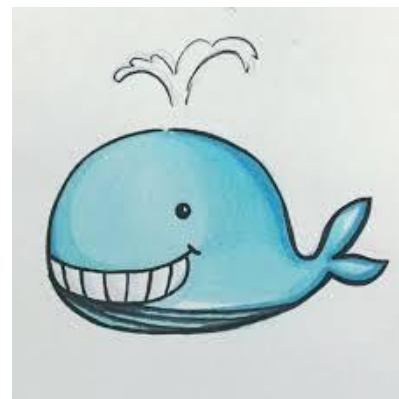


# Question

1. 所有的鯨魚都需要水(T)
2. 所有的魚都需要水。(T)

---

3. 鯨魚是魚。 (F)



Good argument or bad argument ?

# Validity

**Valid argument(對確論證):**

◆ It is impossible that all its premises are true and the conclusion is false.

Conversely,

**Invalid argument:**

◆ It is possible that all the premises are true and the conclusion is false.

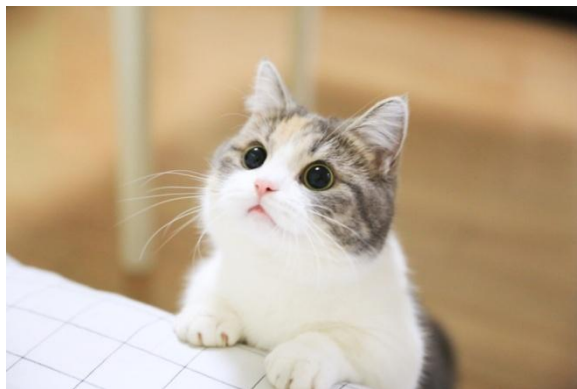
# Case 1a

1. 如果你是中大學生，則你有中大學生証。(T)
  2. 你是中大學生。(T)
  3. 你有中大學生証。(T)
- 

**Valid**

# Case 1b

- 1.所有的哺乳類都是需要空氣生存的生物。(T)
  2. 貓是需要空氣生存生物。(T)
  3. 貓是哺乳類。(T)
- 



**Invalid**

# Case 2

1. 所有的貓都是哺乳類(T)
2. 所有的兔都是哺乳類。(T)

---

3. 所有貓是兔。(F)

**Invalid**



# Case 3a

1. 所有長髮的都是可愛的。(F)
2. 所有可愛的都是生理男性。(F)
3. 圖中的人是長髮的。 (T)

---

4. 圖中的人是生理男性。(T)



Valid



# Case 3b

1. 所有不達156 cm 高的人都是小孩。 (F)
  2. 夏綠蒂是小孩。 (T)
- 
3. 夏綠蒂不達156cm 高。 (T)

**Invalid**





# Case 4a

1. 所有靚仔都有老婆。 (F)

2. 八丙金有老婆。 (T)

---

3. 八丙金是靚仔。 (F)



**Invalid**

# Case 4b

1. 所有不達156 cm 高的人都是小孩。(F)
  2. 陳希敏不達156 cm 高。(T)
- 
3. 陳希敏是小孩。(F)

**Valid**

# Remarks

*An argument is valid or not **does not depend** on the actual truth value of the **premises** and the **conclusion**.*

| Premises | Conclusion | Validity |
|----------|------------|----------|
| T        | T          | ?        |
| T        | F          | Invalid  |
| F        | T          | ?        |
| F        | F          | ?        |

# Special Case 1

1. 陳小波是人。
  2. 陳小波不是人。
- 
3. 鯨魚是魚。

**Valid**

**Premise 1 and Premise 2 is inconsistent. They cannot be both true.**

**Therefore, it is logically impossible that the conclusion is false and all the premises are true**

# Special Case 2

1. The government knows the need of all citizens.
  2. The air pollution problem in Hong Kong is not serious.
- 
3. Mimi has a tail or Mimi does not have a tail.

**Valid**

The conclusion is necessarily true.

Therefore, it is logically impossible that the conclusion false and all premises is true

# Vacuous Validity

- Arguments with **inconsistent premises/ necessarily false premise(s)**
- Arguments with a **necessarily true conclusion**
- **counter-intuitive?**

# Any better accounts ?

**Def<sub>new</sub>:** For any valid argument, it is impossible that all the premises are true and the conclusion is false; and

- 1. it is not the case that (a) the premises are necessarily false or inconsistent , and**
- 2. it is not the case that (b) the conclusion is necessarily true.**



# Problems of the suggested definition

- The concept of validity no longer apply on the mathematical proof / proof of theorem in Logic.
- Against the second additional constraint

# Problems of the suggested definition

1. Hello Kitty is a cat with a month.
2. Hello Kitty does not have a month. (inconsistent Premises)
3. Hello Kitty is a cat.

1. 所有可愛的貓都是貓。
2. 不是貓的都不是可愛的貓。 (Necessarily True)

It will eliminate some cases which we consider as “good” inference.

# Truth Preserving

## We will keep the original definition

- It captures an important feature of good deductive argument – truth preserving
- When we say the argument is invalid, what we means is that it cannot guarantee the truth of the conclusion.

**Remarks: An argument is either valid or invalid.  
We will not say an argument is N % valid.**

# Discussion

1. 獲得我最喜愛男歌手的人都是唱歌好聽的男歌手。
2. 古天樂獲得我最喜愛男歌手。
3. 古天樂是唱歌好聽的男歌手。



# Soundness

$$\text{Sound Argument} = \text{Valid Argument} + \text{All True Premises}$$

## **Sound argument:**

- A deductive argument that is valid and has all true premises.

# Examples

## Example 1

1. 蘇格拉底是人類。 (T)
  2. 所有人類都會死亡。 (T)
- 
3. 蘇格拉底會死亡。 (T)

## Example 2

1. Putin is a bear. (F)
  2. All Bears are mortal. (T)
- 
3. Putin is mortal. (T)

# Reference



- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.



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# **CATEGORICAL SYLLOGISM I**

CHAN, Hei Man

# Validity

**Valid argument(對確論證):**

◆ It is impossible that all its premises are true and the conclusion is false.

Conversely,

**Invalid argument:**

◆ It is possible that all the premises are true and the conclusion is false.

# Question 1



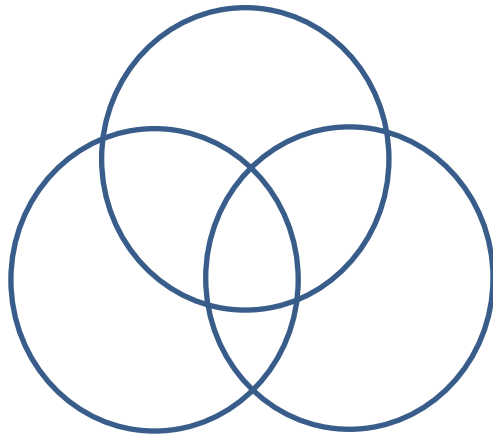
**Someone is sick.**

**Someone is unhappy.**

**Someone is sick and unhappy**

# Any tools for checking validity?

## ➤ Categorical Syllogism



# Agenda



- **Categorical statement (/proposition)**
- **Quality and Quantity**
- **Boolean interpretation Vs Aristotelian interpretation**
- **Venn Diagram**

# Categorical Statement

## Categorical statement

- ◆ A statement that relates two classes, or categories, is called a categorical proposition.
- ◆ The classes in question are denoted respectively by the subject term and the predicate term.

# Examples



1. 靚仔無本心。
2. 俄羅斯人是戰鬥民族。
3. 所有稱為三文魚的魚都可以生食。
4. All cats catch mice.
5. Some coke is designed for people on a diet.



# Standard Categorical Statement

**Components:**

**Quantifier:** *“All” ; “Some” ; “No”*

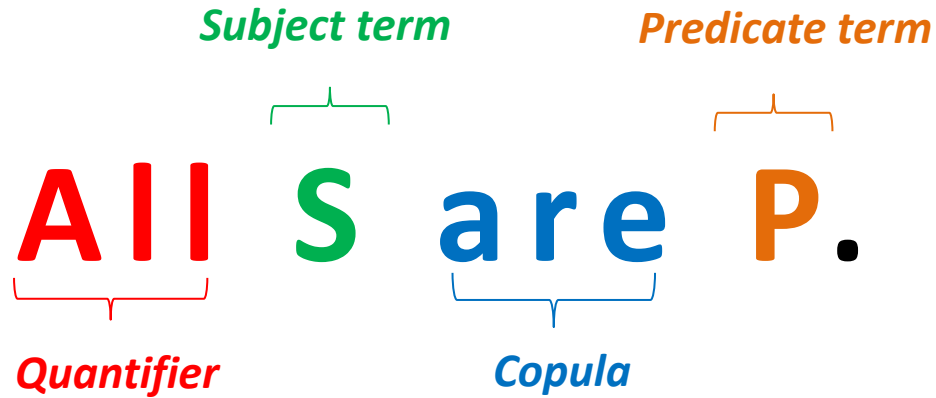
**Subject term**

**Predicate term**

**Copula:** *the words link Subject term and*

*Predicate term, i.e. “are”, “are not”*

# Standard Categorical Statement



# Types of Categorical Statement

- (1) those that assert that the whole subject class is included in the predicate class,
- (2) those that assert that part of the subject class is included in the predicate class,
- (3) those that assert that the whole subject class is excluded from the predicate class, and
- (4) those that assert that part of the subject class is excluded from the predicate class.

# Standard Forms of Categorical statement

**Standard form:**

- (1) All S are P.**
- (2) Some S are P.**
- (3) No S are P.**
- (4) Some S are not P.**

**Not standard:**

**Example**

- 1. All S are not P.**
- 2. Some Not-S are P.**

# Quality and Quantity

## Quality

- affirmative quality:

"All S are P" and "Some S are P"

- negative quality:

"No S are P" and "Some S are not P" .

# Quality and Quantity

## Quantity

◆ **Universal : "All S are P" and "No S are P"**

**e.g. All cats are mammal.**

**No whales are fishes.**

◆ **Particular: "Some S are P" and "Some S are not P"**

**E.g. Some people are bad persons.**

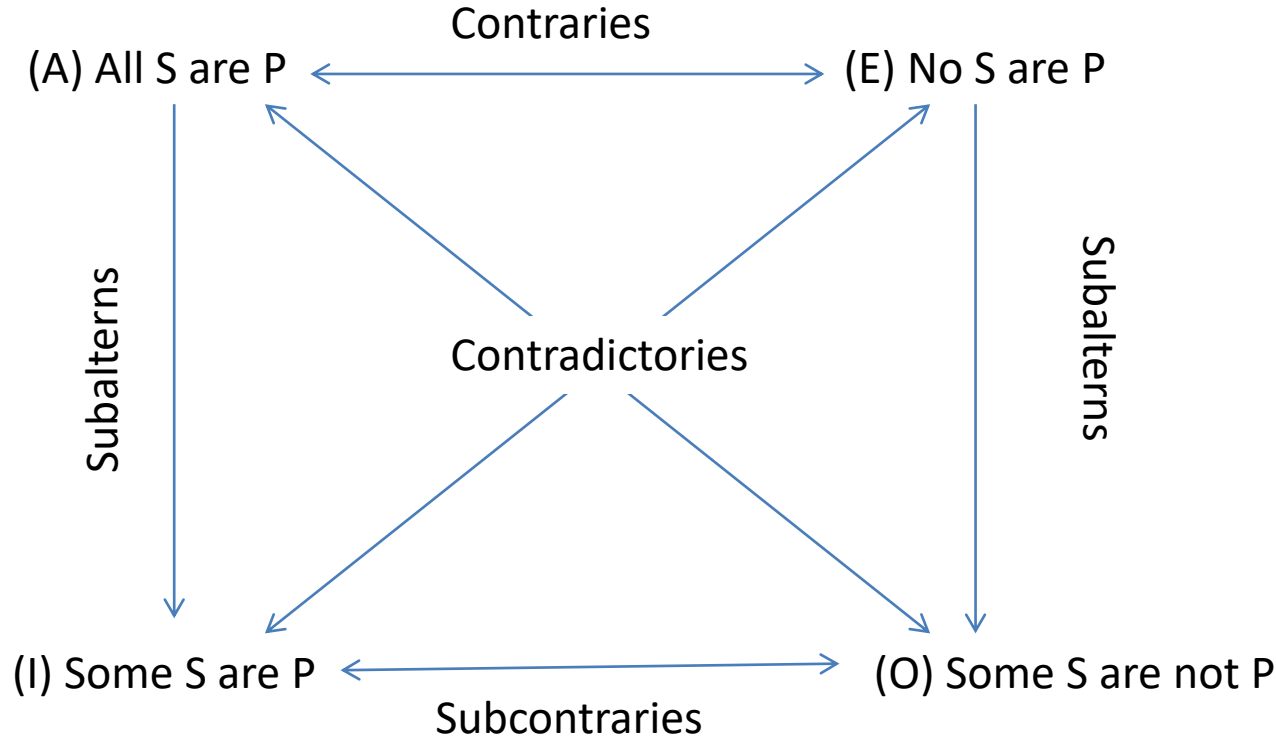
**Some insects are not lovely animals.**

# A,E,I,O

- The universal affirmative is called an **A Proposition**
- the universal negative is called an **E proposition**
- the particular affirmative is called an **I proposition**
- the particular negative is called an **O proposition**.

| Proposition       | Letter Name | Quantity   | Quality     |
|-------------------|-------------|------------|-------------|
| All S are P.      | A           | Universal  | Affirmative |
| No S are P.       | E           | Universal  | Negative    |
| Some S are P.     | I           | Particular | Affirmative |
| Some S are not P. | O           | Particular | Negative    |

# The traditional square of opposition





# The traditional square of opposition

| Relation       | Definition                                 |
|----------------|--------------------------------------------|
| Contradictory  | opposite truth value                       |
| Contrary       | at least one is false (not both true)      |
| Subcontrary    | at least one is true (not both false)      |
| Subalternation | truth flows downward; falsity flows upward |

# The traditional square of opposition

- “All S are P”  $\rightarrow$  “Some S are P”
- “No S are P”  $\rightarrow$  “Some S are not P”
- “Some S are P” and “Some S are not P” can not both be false.
  - There is at least one S, so this S is either P or not P.

# Example

**All cats have a tail.**



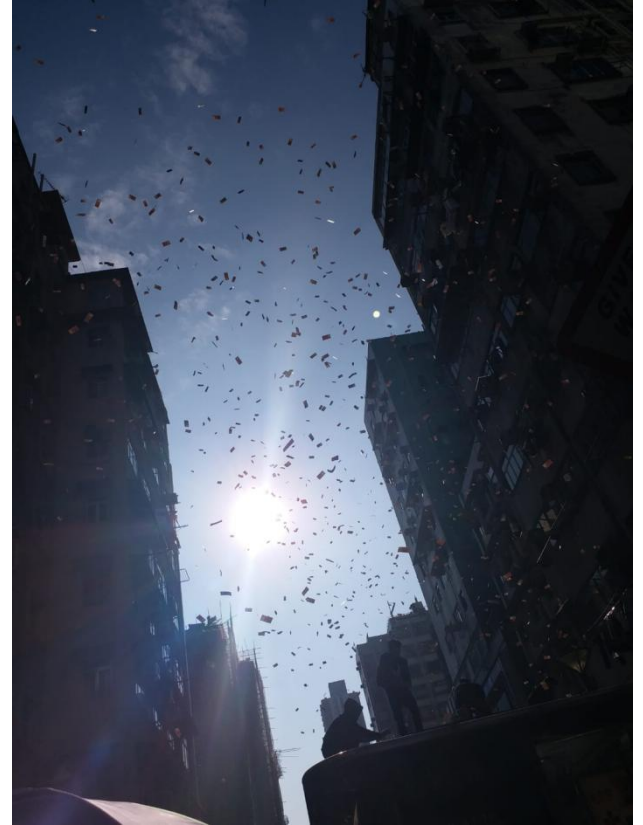
**Some cats have a tail.**



# Scenario 1

Imagine someone told you that  
“Come! All banknotes on the floor of  
NAH11 are welcome to be collected”

However, you found no banknotes  
when you come to class.



# Scenario 2



Imagine a lecturer told you that  
“All students who fail the course are who  
need to retake the course.”

Do the lecturer presuppose there is at  
least one student need to retake?

# Existential import



For **natural language**:

- We have existential import for some “all”-statement
- Depend on context!

How about formal language?

# Aristotelian Standpoint

**Aristotelian: *“All” implies “some”***

“All students who fail the course are who need to retake the course.”

**Implies**

“Some students who fail the course are who need to retake the course.”



# Interpretations

All dragons have wings.



Some dragons have wings.



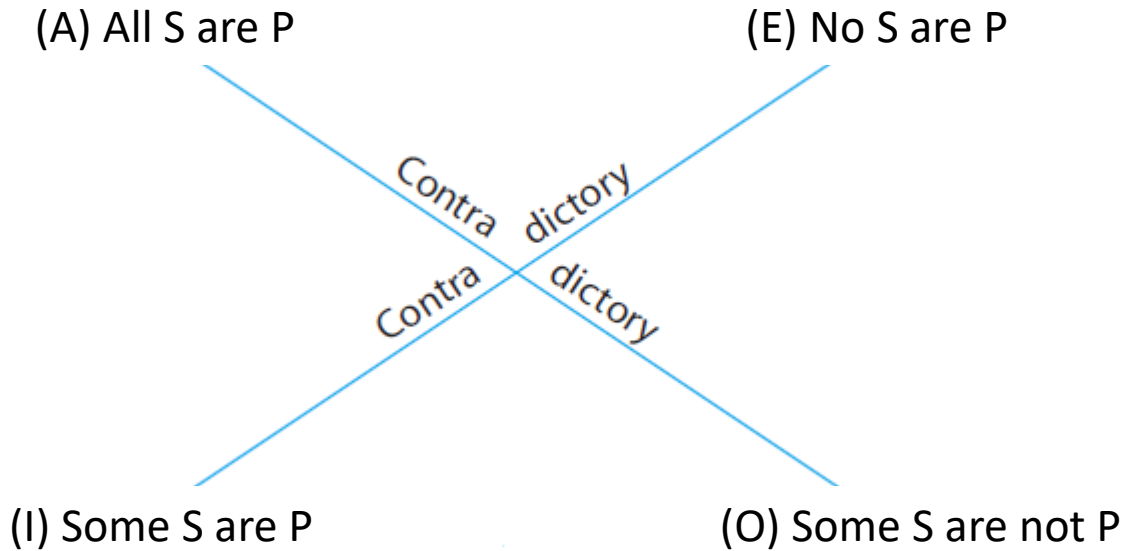
所有獨角獸有角。



有些獨角獸是有角。



# Modern Square



# Boolean Standpoint

**Boolean:**

“All students who fail the course are who need to retake the course.”

**doesn't imply**

“Some students who fail the course are who need to retake the course.”

“If there is such a thing as a student who fails the course , it is a person who need to retake the course.”

# Existential Fallacy

1. All cats are animals.
2. Some cats are animals.

**Boolean: Invalid, existential fallacy**  
**Aristotelian: Valid**

**No existential import  
for empty terms for  
Aristotelian standpoint!**

1. All unicorns are animals.
2. Some unicorns are animals.

**Boolean: Invalid, existential fallacy**  
**Aristotelian: Invalid, existential fallacy**

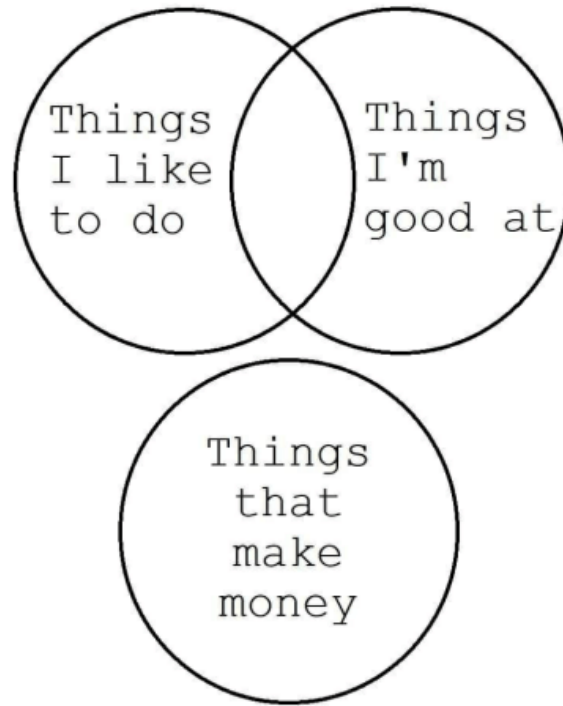
# Aristotelian Vs Boolean

## For non-empty terms

|              | Existential import<br>in<br>"All" / "No"<br>statement | Existential import<br>in<br>"Some" statement |
|--------------|-------------------------------------------------------|----------------------------------------------|
| Aristotelian | Yes                                                   | Yes                                          |
| Boolean      | No                                                    | Yes                                          |

# How to represent a categorical statement?

Venn Diagram of my life:

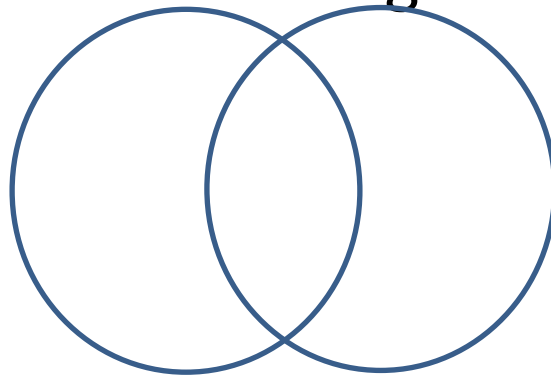


# Venn Diagram

**Represent the relationship of two terms within a categorical statement.**

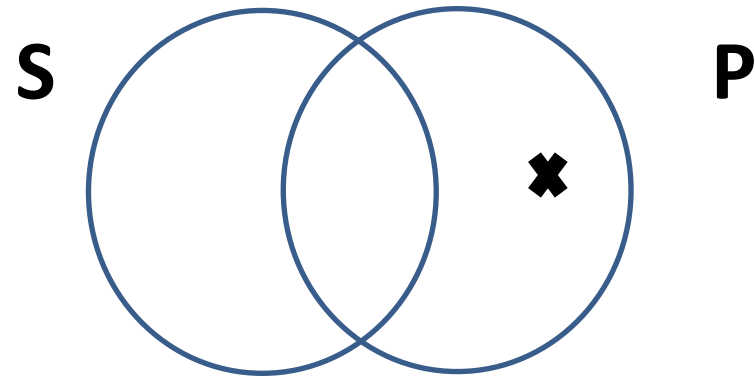
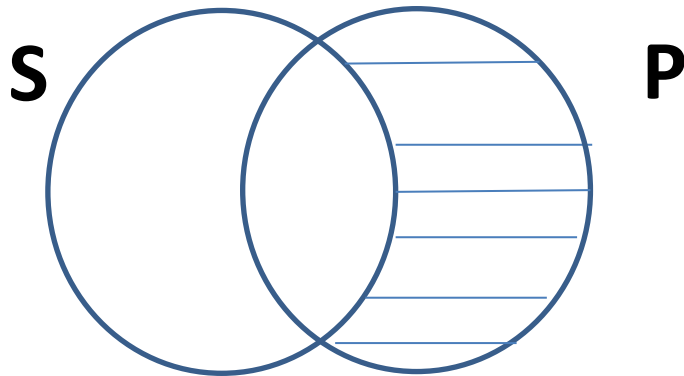
How to represent the four kinds of categorical proposition on a Venn Diagram

- Start with two intersecting circle as follows



# Venn Diagram

- Two kinds of marks are used: shading an area and placing an "X" in an area.
  - Shading an area means that the shaded area is empty,
  - placing an "X" in an area means that at least one thing exists in that area.



# Drawing Venn Diagram

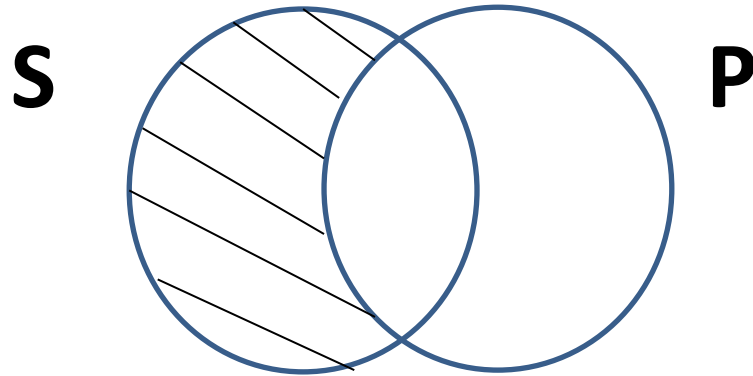
---

- Shading an area means that the shaded area is empty
  - For **universal** statements (*“all” or “no”* )
- Placing an "X" in an area means that at least one thing exists in that area
  - For **particular** statements (*“some”*)



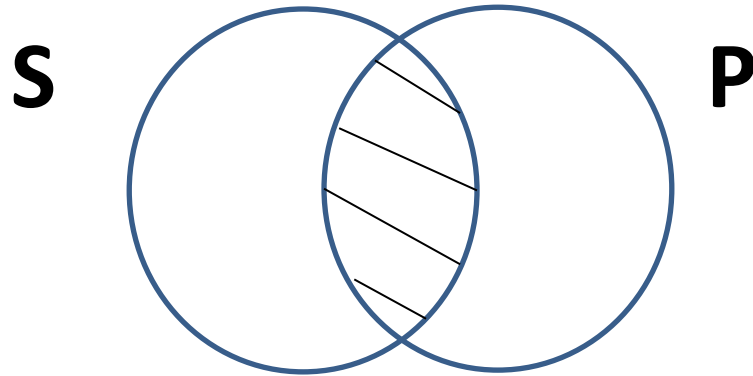
# Venn Diagram

All S are P.



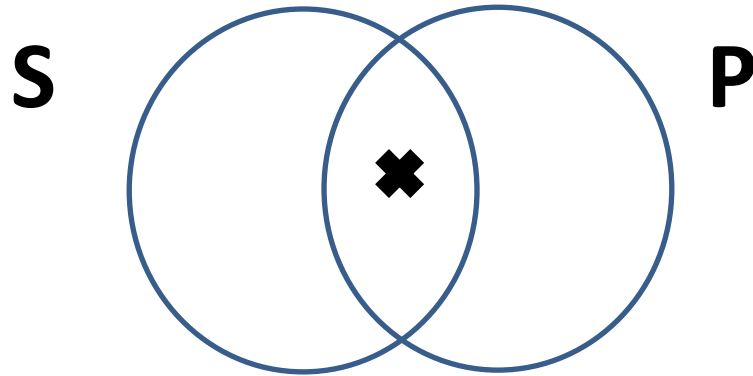
# Venn Diagram

**No S are P.**



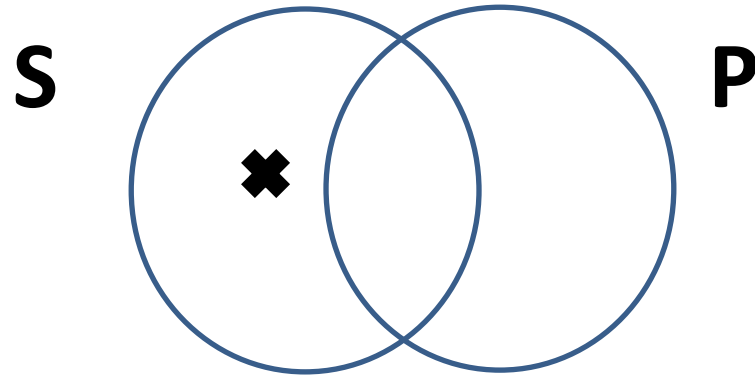
# Venn Diagram

**Some S are P.**

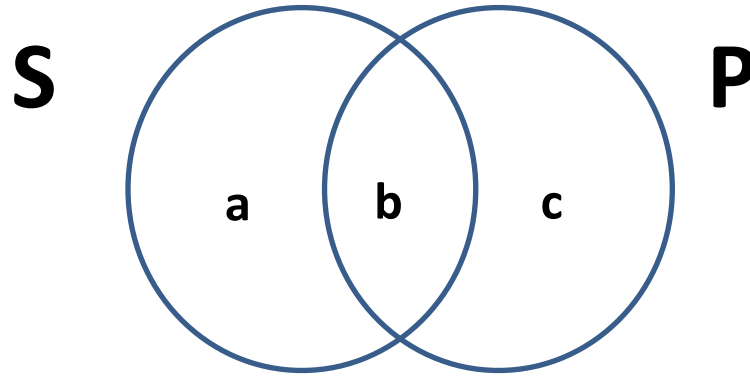


# Venn Diagram

**Some S are not P.**



# Venn Diagram



**A: Shade a**

**B: Shade b**

**C: Shade c**

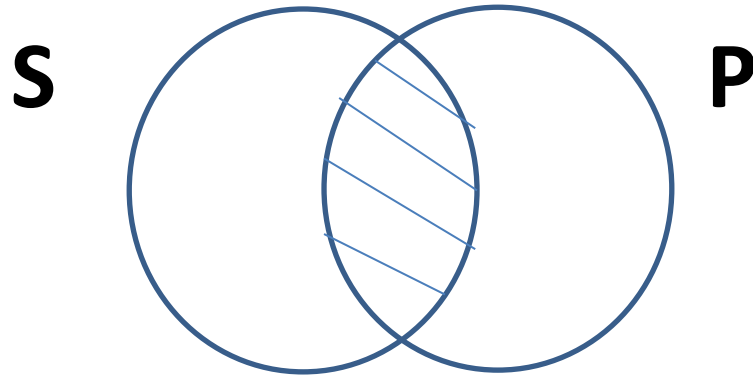
**D: Mark X on a**

**E: Mark X on b**

**F: Mark X on c**

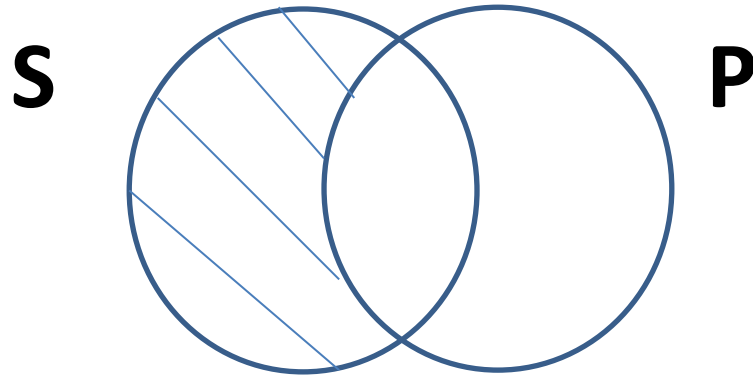
# Discussion 1

All S are not P.  
= No S are P



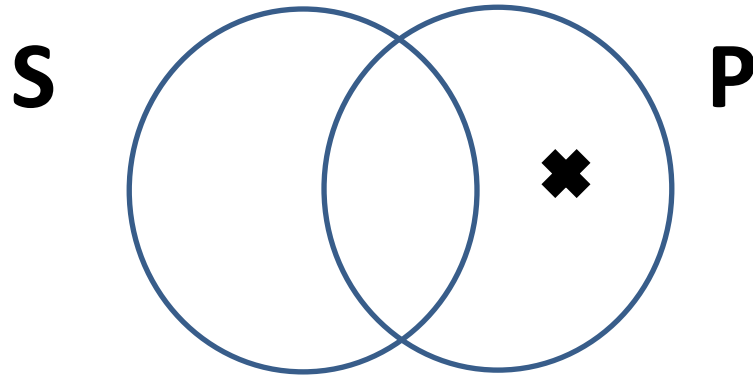
# Discussion 2

No S are not P.  
= All S are P.



# Discussion 3

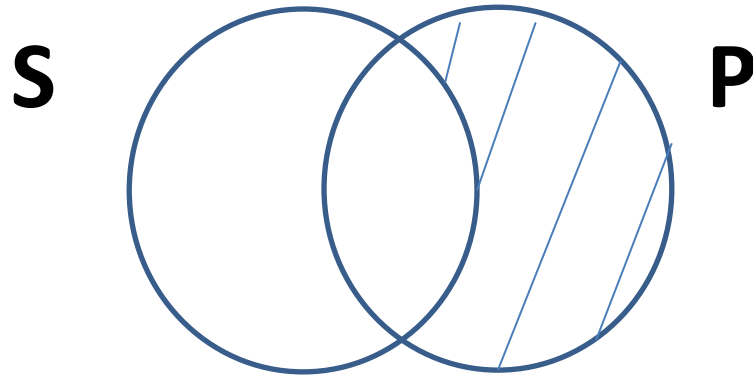
Some non-S are P.  
= Some P are not S.





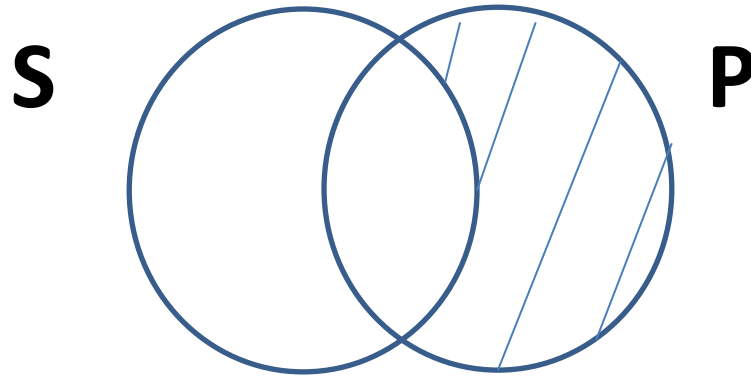
# Discussion 4

All non-S are not P.  
= All P are S.



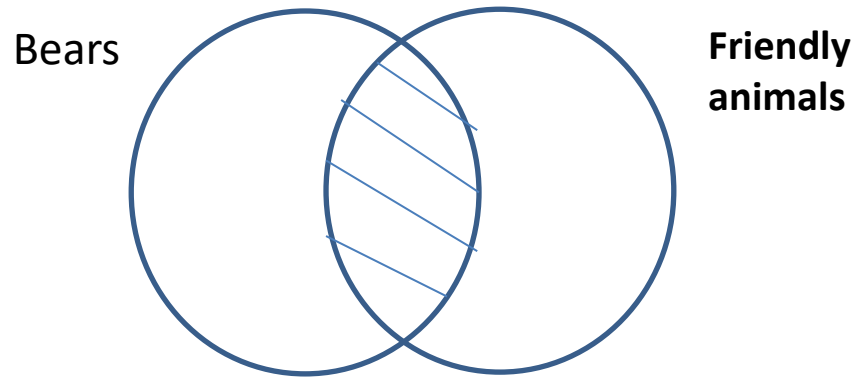
# Discussion 5

It is not the case that Some P are not S .  
= All P are S.



# Discussion 6

No bears are friendly animals.



# Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.

**UGED1111 Logic**

# **CATEGORICAL SYLLOGISM II**

CHAN, Hei Man

# Agenda

---

- **Test the validity with Venn diagram**
- **Translating Ordinary Language Statements into Categorical Form**

# Categorical Syllogism

---

**A CATEGORICAL SYLLOGISM is a**  
*deductive argument* with:

- a. 3 categorical propositions: two premises and one conclusion.**
- b. 3 terms, each occurring twice .**
- c. Each term occurring in two statements.**

# Categorical Syllogism

---

- The MAJOR TERM is the predicate term (P) of the conclusion.
- The MINOR TERM is the subject term (S) of the conclusion.
- The MIDDLE TERM is the term (M) occurring in both premises but not in the conclusion.
- *The MAJOR PREMISE is the premise containing the major term.*
- *The MINOR PREMISE is the premise containing the minor term.*



# STANDARD FORM

---

## STANDARD FORM:

1. Major Premise
  2. Minor Premise
  3. Conclusion
- 

## Example

1. All M are P.
  2. Some M are S.
  3. All S are P.
-

# Check the Validity I

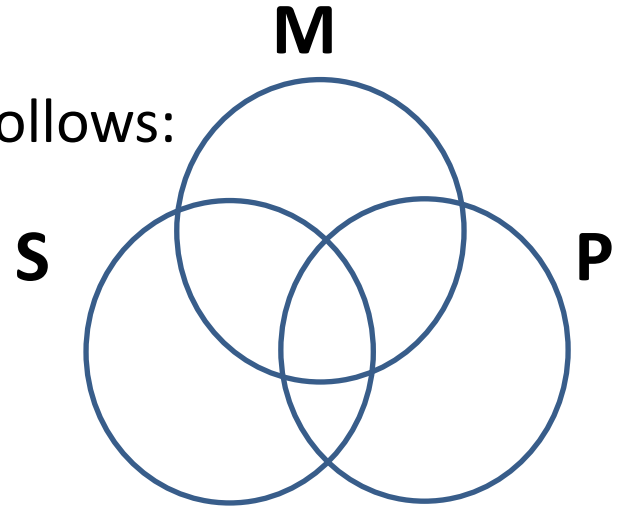
We can check the validity of Categorical Syllogisms with Venn diagrams

1. Draw three circles and label them as follows:

P is the Major term.

M is the Middle term,

S is the Minor term,

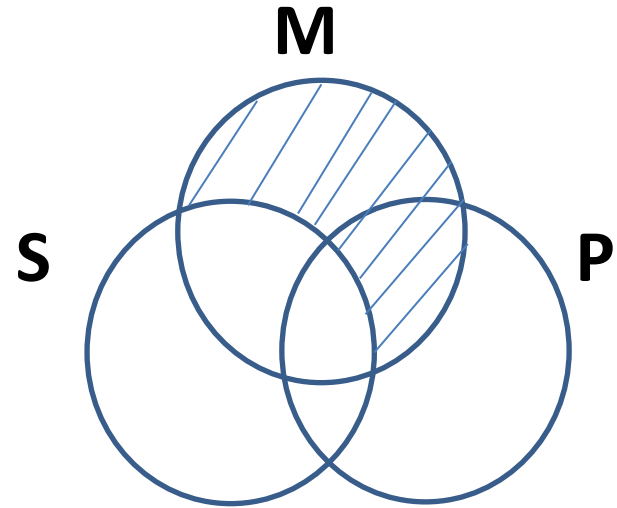


# Check the Validity II

2. Marks (shading or placing an X) are entered only for the premises. No marks are made for the conclusion.

3. **Universal premises** should be marked on the diagram **first** and then particular premises. No priority of premises marking if both premises are universal or both are particular.

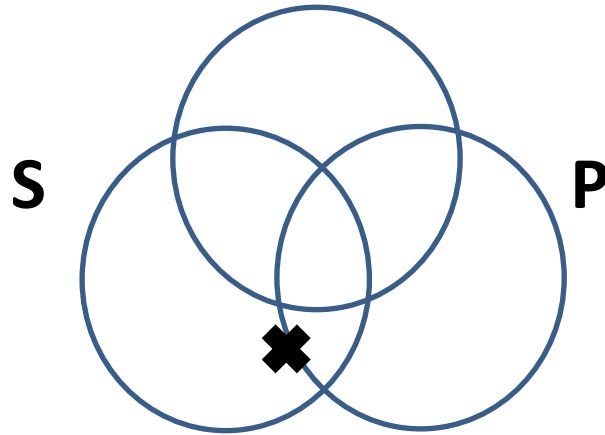
4. Shade the area where no objects can exist for the universal premises



e.g. All M are S.

# Check the Validity III

5. Put an X in the area where at least one object exists. If it is possible to put an X in two areas, then put it on the borderline between the two regions **M**



e.g. Some S are not M.

6. Check if the conclusion is represented on the diagrams. If Yes, then the syllogism is **VALID**. If No, it is **INVALID**.

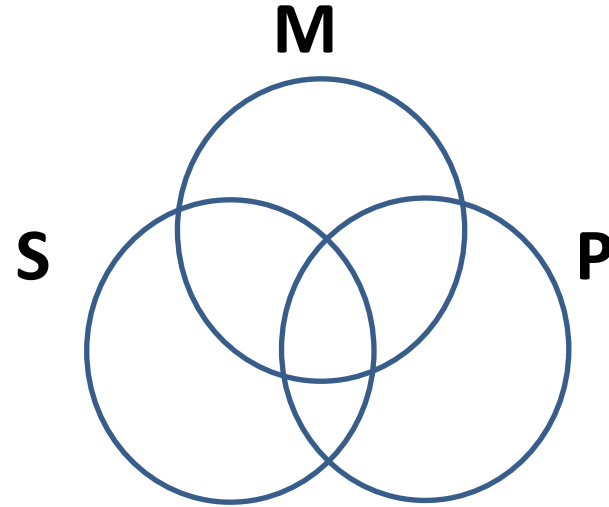
# Recap

## STANDARD FORM:

1. Major Premise

2. Minor Premise

3. Conclusion



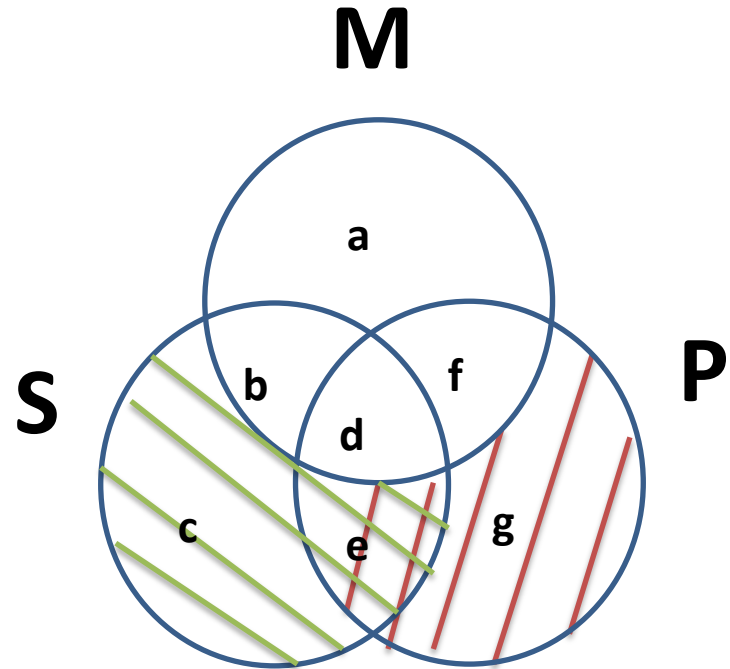
\*Mark Universal premises on diagram first, then particular premises.

\*\* No marks are made for the conclusion.

# Example 1

1. All P are M.
  2. All S are M.
  3. No S are P.
- 

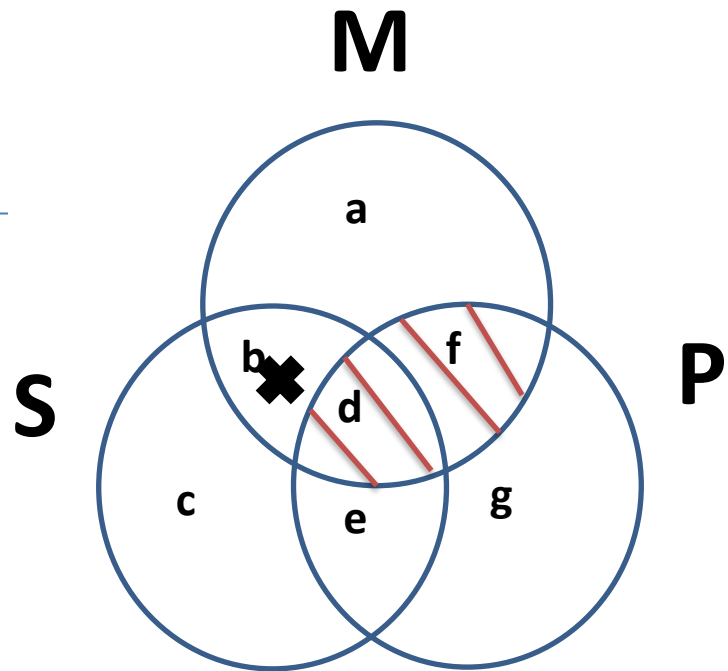
**Invalid**



# Example 2

1. No P are M.
  2. Some S are M.
  3. Some S are not P.
- 

Valid



# Existence

---

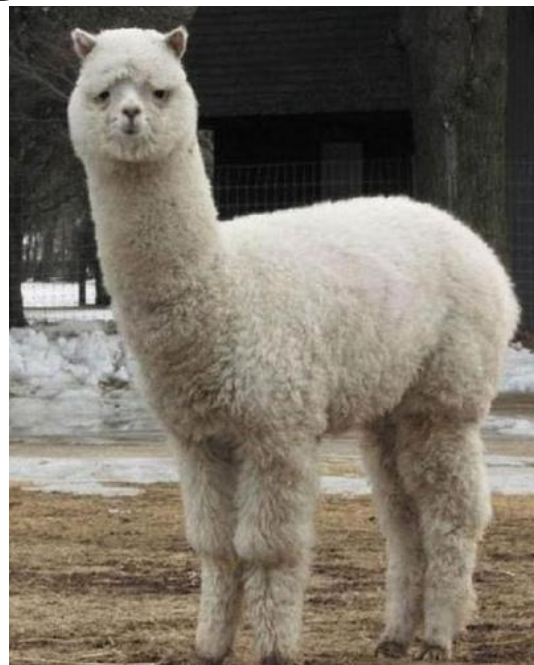
**Some : there exists/is at least one**

**What does it mean by “exist”?**

**Sheep? IRONMAN?**

**草泥馬? Unicorn?**

**Round-square?**

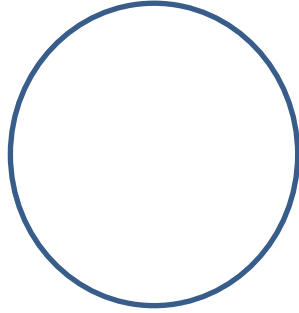




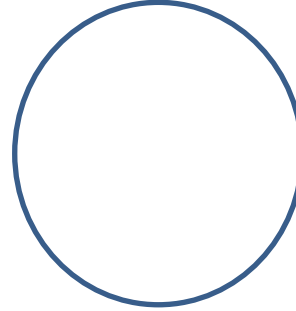
# Existence

---

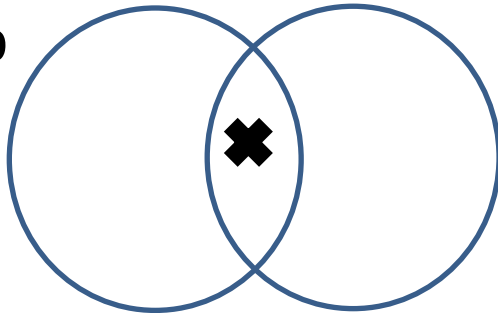
Sheep



Black  
objects



Sheep



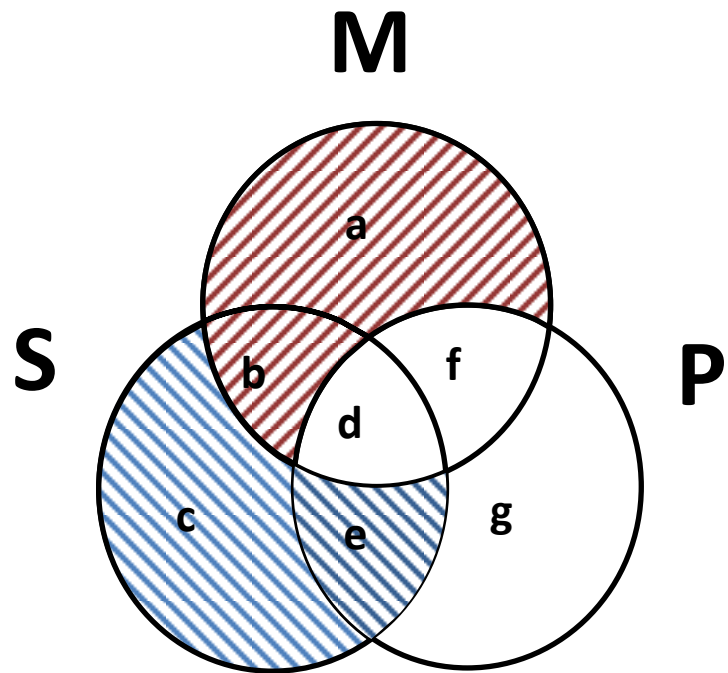
Black  
objects

Some Sheep are black objects.

# Discussion 1

Determine whether the following argument is valid or not

1. All M are P.
  2. All S are M.
- 
3. All S are P.



Valid

# Discussion 2

Determine whether the following argument is valid or not

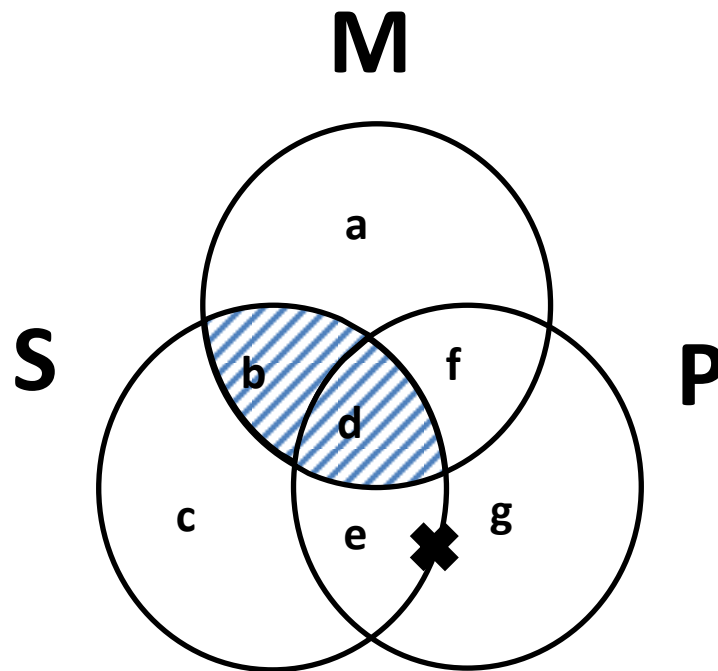
1. Some P are not M.

2. No S are M.

---

3. Some S are not P.

Invalid



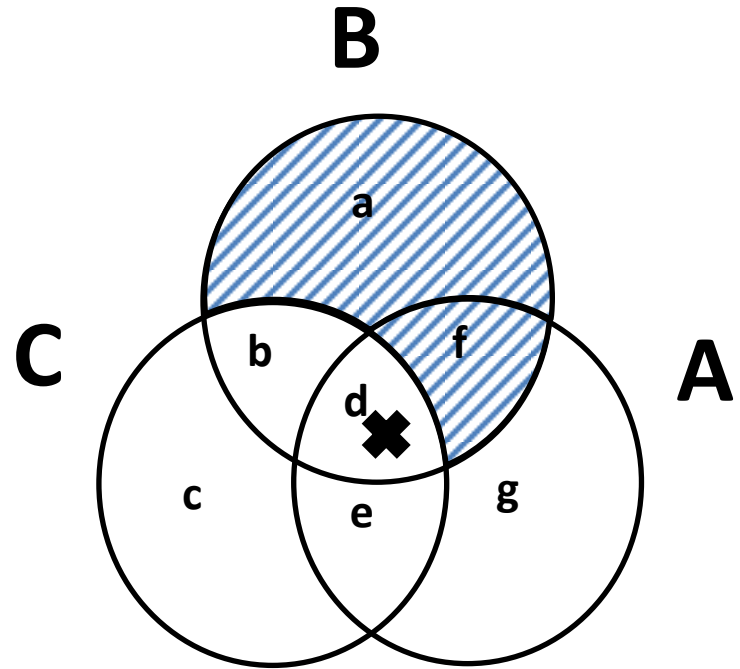
# Discussion 3

Determine whether the following argument is valid or not

No non-C are B. Not all A are not B  
Therefore, some C are A.

1. Some A are B.
2. All B are C.
3. Some C are A.

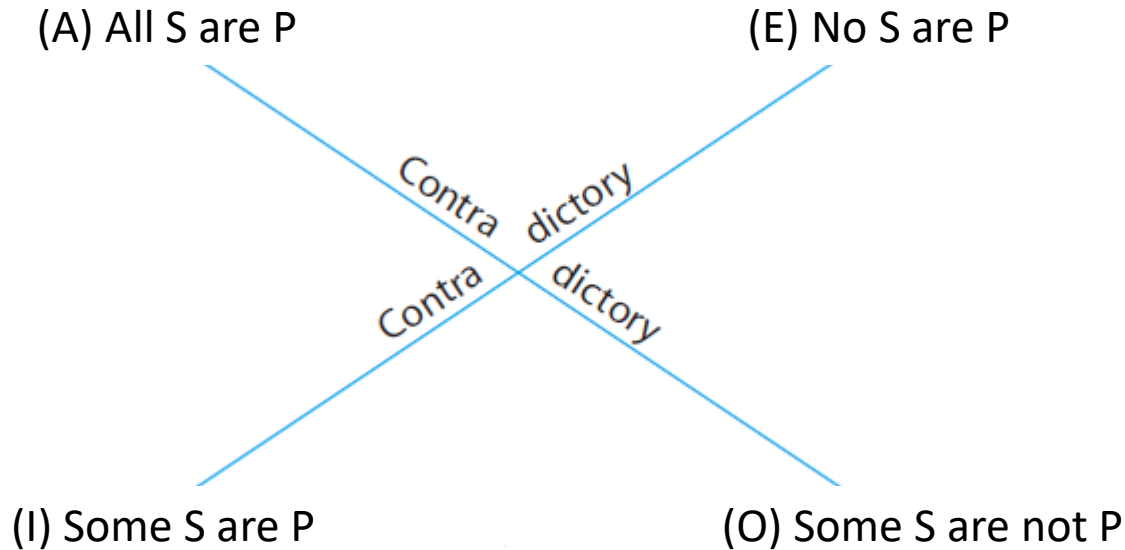
Valid



# Recap

---

## Modern Square



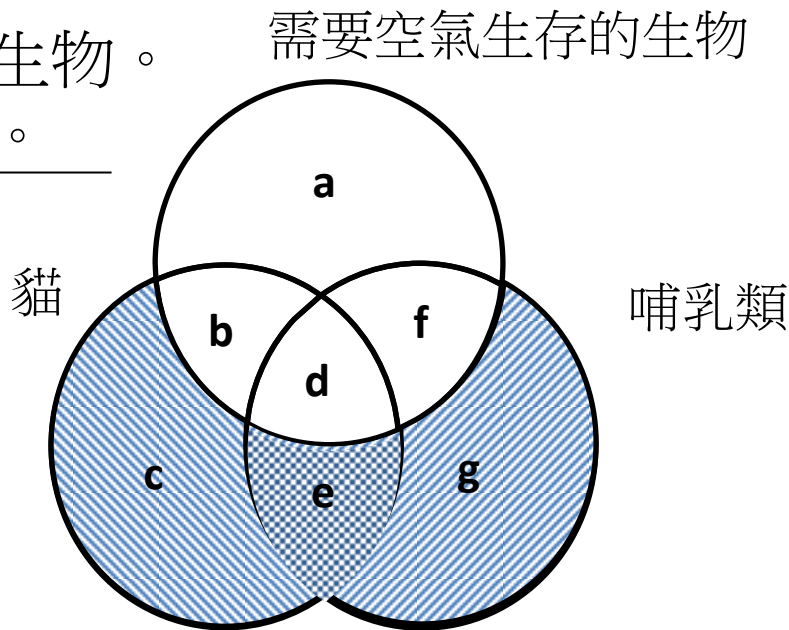
**\*\* “All” does not imply “some”**

# Example

Determine whether the following argument is valid or not

1. 所有哺乳類都是需要空氣生存的生物。
  2. 所有貓都是需要空氣生存的生物。
- 
3. 所有貓都是哺乳類。

Invalid



# Translation

---

## General rule

• Understand the meaning of the given statement, and then re-express it in a new statement that has a **quantifier, subject term, copula, and predicate term.**

Standard quantifier: All ; No; Some

Copula: are; are not

Subject term and predicate term must be a noun.

# Terms Without Nouns

---

If a term consists of only an adjective, a noun should be introduced to make the term genuinely denotative.

Example:

“All swans are **white**.”

- “All swans are **white animal**.”



# Nonstandard Verbs

---

The only copulas that are allowed in standard-form categorical propositions are "are" and "are not."

Statements in ordinary usage often incorporate other forms of the verb "to be."

Example:

"Some staffs **will get year-end bonus.**"

- "Some staffs **are persons who will get year-end bonus.**"

# Singular Proposition

---

**Categorical logic concerns categories.**

How about the proposition like:

"Socrates is mortal"

- "All persons identical to Socrates are persons who are mortal."

**Singular propositions are typically translated into universal by means of a parameter.**

**Why do we use “all” instead of “Some” as in such cases?**

# Adverbs and Pronouns

---

- When a statement contains a **spatial** adverb such as "where," "wherever," "anywhere," "everywhere," or "nowhere," or a **temporal** adverb such as "when," "whenever," "anytime," "always," or "never,"

It may be translated in terms of "**places**" or "**times**" respectively.

# Adverbs and Pronouns

---

**Statements containing pronouns such as "who", "whoever", "anyone", "what" "whatever" or "anything" may be translated in terms of "persons" or "things" respectively.**

Examples:

1. "Whoever works hard will succeed."
  - "All persons who work hard are persons who will succeed."
2. "Nowhere are good without logic."
  - "No good places are places without logic."

# Unexpressed Quantifiers

---

**In introducing the quantifiers one must be guided by the most probable meaning of the statement.**

Example:

1. "There are monkeys in CUHK."
  - "Some monkeys are animals in CUHK."
2. "A few lecturers are nice."
  - "Some lecturers are nice persons."

# Conditional Statements

---

When the antecedent and consequent of a conditional statement talk about the same thing, the statement can usually be translated into categorical form. Such statements are always translated as universals.

“If a person drinks red bull, then he will have wings.”

- “All persons who drink red bull are persons who have wings.”

**‘only if ’**

“An animal is a fish only if it lives in water.”

- “All fishes are animals that live in water.”

# “Unless”

---

**“Unless” means if not**

*“Tomatoes are edible unless they are spoiled.”*

- “All unspoiled tomatoes are edible tomatoes.”

*“Students will fail this course unless they attend the exam.”*

- “All students who do not attend the exam are students who fail this course.”

**Statements containing “unless” are translated as categorical propositions having negated subject terms.**

\*Usually we avoid negative terms

*“Unless the sun is shining, a rainbow cannot occur.”*

- “All times that a rainbow occurs are times that the sun is shining.”

# Exclusive Propositions

---

**Many propositions that involve the words "only," "none but," "none except," and "no ...except" are exclusive propositions.**

## Example

“Only nuclear-powered vehicles are suitable for deep-space exploration.”

- “All vehicles which are suitable for deep-space exploration are nuclear-powered vehicles .”



# Exclusive Propositions

---

- “None but hard-working students will attend 8:30 lessons.”  
“All students who attend 8:30 lessons are hard-working students “

**language following "only," "none but " "none except," and "no ... except" goes in the predicate term of the categorical proposition.**

# “The Only”

---

**Propositions that involve the words “the only,” are different from “only”**

Example

“The only mammals which lay eggs are Platypus”.

- “All egg-laying mammals are Platypus.”

**Language following “the only” goes in the subject term of the categorical proposition.**

# Pragmatic Meaning

---

## Semantics Vs Pragmatic

“你的桌上是不是有胡椒粉?”

**Context- dependent**

**\*For exam, we translate the statement based on the literal meaning only. For daily usage , we should also understand the meaning under that particular context.**

# Discussion 4



1. **“Not everyone who celebrate Valentine is in love.”**  
**“Some persons who celebrate Valentine are not persons in love.”**
2. **“No birds excepts penguin can swim.”**  
**“All birds that can swim are penguins.”**

# Discussion 4

**3. “It is not the case that nobody loves geeks.”**

**“Some persons are persons who loves geeks.”**

**4. “An argument is not sound unless it is valid”**

**“All sound arguments are valid arguments”**

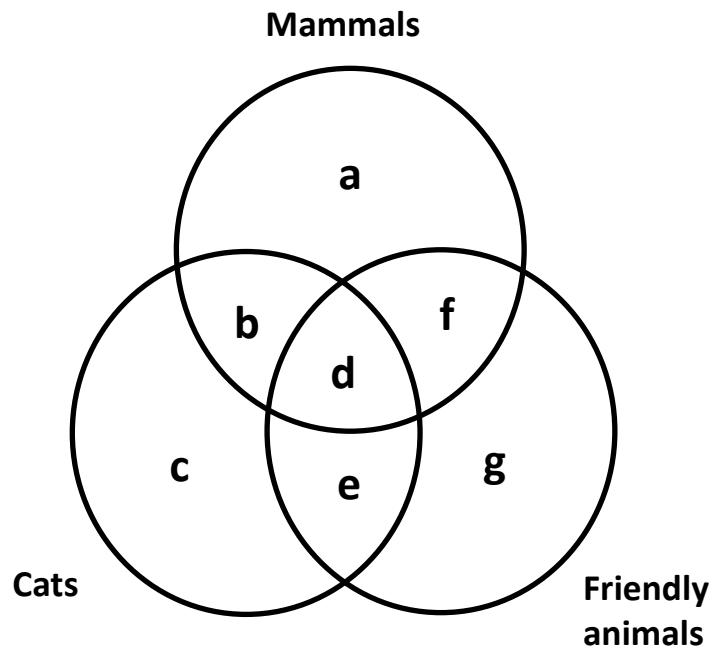
**5. “No Hong Kong people will get married except who have owned a flat.”**

**“All Hong Kong people who get married are Hong Kong people who have owned a flat. “**

# Discussion 5

If it is not a mammal, it is not a cat. A few mammals are friendly. Therefore, Some cats are friendly animals.

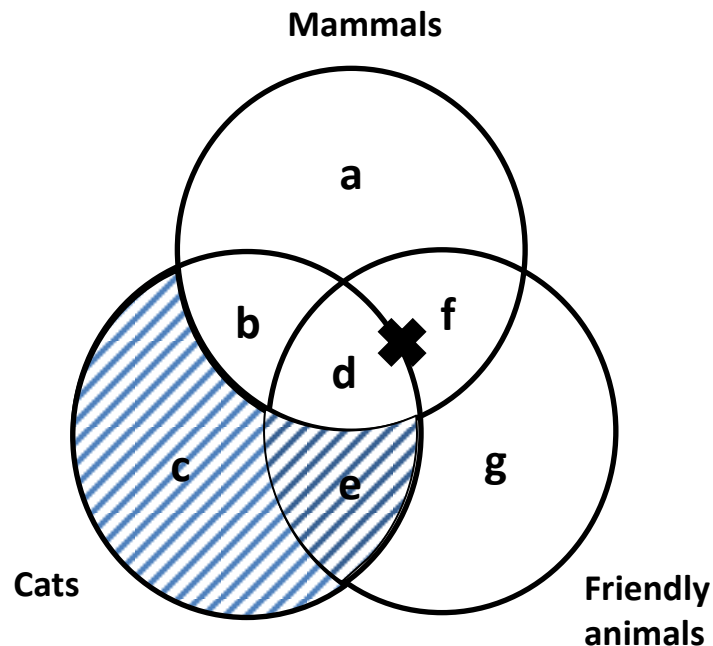
1. Some mammals are friendly animals.
  2. All Cats are mammals.
- 
3. Some cats are friendly animals.



# Discussion 5

1. Some mammals are friendly animals.
  2. All Cats are mammals.
- 
3. Some cats are friendly animals.

Invalid



# Discussion 6

---

**It is false that everyone in love like flowers. Only who like flowers will receive flower. Therefore, it is not the case that everyone in love will receive flowers.**

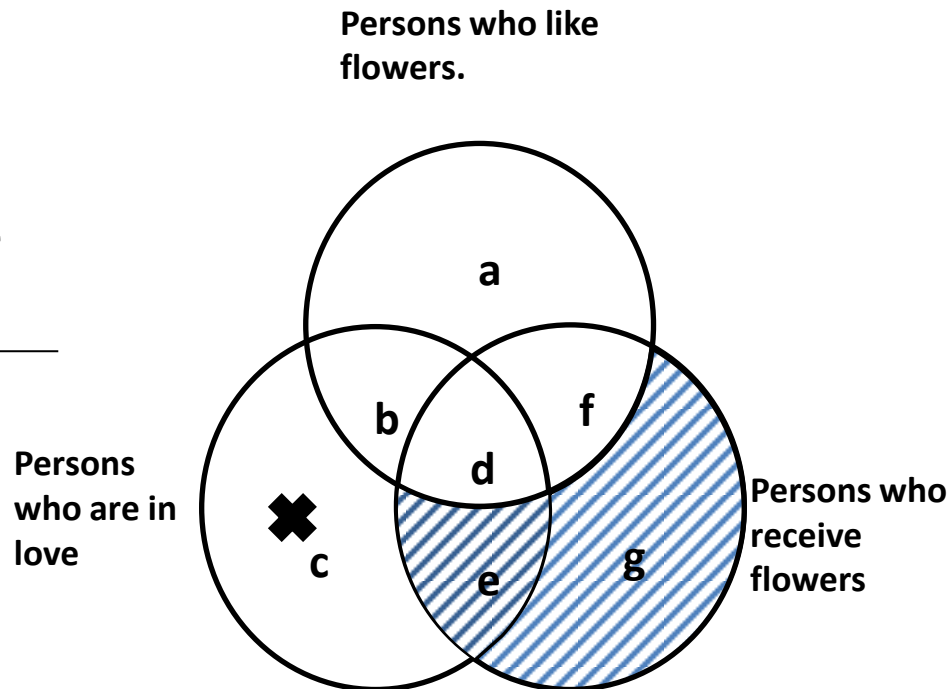
- 1. All person who receive flowers are persons who like flowers.**
- 2. Some persons in love are not persons who like flowers.**
- 3. Some persons in love are not persons who receive flowers**



# Discussion 6

1. All persons who receive flowers are persons who like flowers.
  2. Some persons who are in love are not persons who like flowers.
- 
3. Some persons who are in love are not persons who receive flowers

Valid



# Translation in Categorical Logic

Singular propositions are typically translated into universal by means of a parameter.

Why do we use “all” instead of “Some” as in such cases?

Consider , “Socrates is handsome.”

If we try to translate it into a “some “ statement, then it will be “Some persons are persons who are handsome”

Does this translation express the meaning of “Socrates is handsome.” better than "All persons identical to Socrates are persons who are handsome." ?

# Translation in Categorical Logic



**Even though we try our best to translate in a way to keep the original meaning of statements in ordinary language, sometimes it is hard for us to retain the meaning of the original statement.**

# Case

## Consider the validity of the following arguments:

- |                                         |                            |
|-----------------------------------------|----------------------------|
| 1. 如果上帝無法造出自己舉不起來的石頭，<br>上帝不是全能的。       | 1. $\sim P \supset \sim Q$ |
| 2. 如果上帝可以造出自己舉不起來的石頭，<br>上帝不是全能的。       | 2. $P \supset \sim Q$      |
| 3. 上帝可以造出自己舉不起來的石頭或上<br>帝無法造出自己舉不起來的石頭。 | 3. $P \vee \sim P$         |
| <hr/>                                   |                            |
| 4. 上帝不是全能的。                             | // $\sim Q$                |

# Problem of Categorical Logic

**Some good arguments in natural language can not be proved in categorical logic.**

- 1. All cats are mammals.**
  - 2. All cats' tails are mammals' tails.**
- 

We will consider it as a good argument. However, we cannot prove it in the system of Categorical Logic.

# Problem of Categorical Logic

1. All cats are mammals.
  2. All cats' tails are mammals' tails.
- 

Categorical logic can only handle arguments with 3 terms.  
**However, the argument involves 4 terms.**

**The expressive power of categorical logic is low.**

# Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.

UGED1111 Logic

Symbolic Language  
and  
Truth Table



# Symbolic language



- To show the logical structure of an argument
- To help checking the validity of an argument

# Propositional Logic

- Propositions (/statements)

- $P, Q, R, S, \dots$

- Logical operators(/connectives)

- $\sim, \vee, \cdot, \supset, \equiv$

# Truth Function



A truth function is a function that takes truth values as input and generate a truth value as output.

**Propositional Logic is a truth functional system.**

The truth function is any compound proposition whose truth value is completely determined by the truth values of its components.

# Simple statement

**A simple statement /atomic statement is one that does not contain any other statement as a component.**

Examples:

1. Human activity is the main reason of global warming.
2. 教育局禁止學生會舉辦丘比特活動。
3. 我在NA CAN 吃午餐。
4. 中華白海豚的棲息地不停被破壞。
5. Judy attends today's class.

# Compound statement

A compound statement (複合句) is one that contains at least one simple statement as a component.

Example:

1. **It is not the case that** human activity is the main reason of global warming.
2. 明愛禁止學生會舉辦丘比特活動，而且這事被廣泛報導。
3. 我在NA CAN 吃午餐，**或者**我在MED CAN 吃午餐。
4. **如果**香港繼續填海，則中華白海豚的棲息地不停被破壞。
5. Judy will attend today's class **if and only if** Tiffany will attend today's class.

# Operators

**Logical operators:**  $\sim$ ,  $\vee$ ,  $\cdot$ ,  $\supset$ ,  $\equiv$

**Operators you have used in daily life:**  $+$ ,  $-$ ,  $\times$ ,  $\div$

➤  $(x + y)$ ,  $(x - y)$ ,  $(x \times y)$ ,  $(x \div y)$

□  $\sim(P \equiv (Q \vee R)) \supset (\sim R \cdot S)$

□  $[9 + (2 - 1) \times 6] \div 3$

# Logical Operators

| Operator  | Name       | Logical function      | Used to translate               |
|-----------|------------|-----------------------|---------------------------------|
| $\sim$    | Tilde      | Negation (否定句)        | Not, it is not the case that... |
|           |            |                       | 不是，不，並非...                      |
| $\cdot$   | Dot        | Conjunction (合取句)     | And, also, but...               |
|           |            |                       | 及，而且，但是...                      |
| $\vee$    | Wedge      | Disjunction (分取句)     | ...or...                        |
|           |            |                       | 或                               |
| $\supset$ | Horseshoe  | Conditional (條件句)     | If...then, only if...           |
|           |            |                       | 如果...則，僅當...                    |
| $\equiv$  | Triple bar | Bi-conditional (雙條件句) | If and only if                  |
|           |            |                       | 當且僅當                            |

# Negation

To translate “not”.

Example:

Snow is white. (**P**)

Snow is not white. ( $\sim$ **P**)

| <b>p</b> | <b><math>\sim</math>p</b> |
|----------|---------------------------|
| <b>T</b> | <b>F</b>                  |
| <b>F</b> | <b>T</b>                  |



# Negation



What is the negation of this statement?

“Bella likes expensive dresses.”

To properly express its negation:

“It is false that Bella likes expensive dresses.”

“It is not the case that Bella likes expensive dress.”

# Conjunction

To translate “and”

Example:

Both *Ironman* and *Ant-man* are  
Marvel Characters. ( $P \cdot Q$ )

| p | q | $p \cdot q$ |
|---|---|-------------|
| T | T | T           |
| T | F | F           |
| F | T | F           |
| F | F | F           |

- *Ironman* is a Marvel Character. (P)
- *Ant-man* is a Marvel Character. (Q)

# Disjunction

To translate “or”

我上課或者我睡覺。 ( $P \vee Q$ )

- 我上課 ( $P$ ) ; 我睡覺 ( $Q$ )

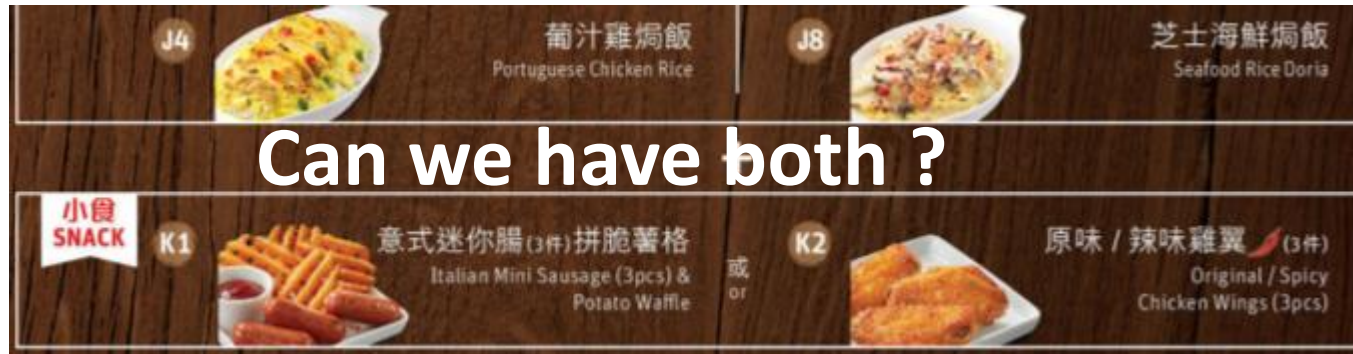
Either you are clever or you will fail the course.

( $M \vee N$ ) ?

- You are clever. ( $M$ )
- You will fail the course. ( $N$ )

| $p$ | $q$ | $p \vee q$ |
|-----|-----|------------|
| T   | T   | T          |
| T   | F   | T          |
| F   | T   | T          |
| F   | F   | F          |

# Disjunction



Example :You're either with us, or against us.

| Inclusive-or | Exclusive-or                       |
|--------------|------------------------------------|
| $p \vee q$   | $(p \vee q) \cdot \sim(p \cdot q)$ |

# Conditional

To translate “if...then”

If Joe feels bored, he will read books about logic. ( $P \supset Q$ )

- Joe feels bored. ( $P$ )
- Joe will read books about logic. ( $Q$ )

| p | q | $p \supset q$ |
|---|---|---------------|
| T | T | T             |
| T | F | F             |
| F | T | T             |
| F | F | T             |

# Conditional

$P \supset Q$

If P, then Q

“P” is antecedent (前項/前件).

“Q” is consequent (後項/後件).

**Material conditional**

**The whole conditional is false when and only when the antecedent is true and the consequent is false.**

# Conditional

**Counter-intuitive ?**

**Why the conditional is true if the antecedent is false?**

**It seems that we won't accept “如果地球是正方體，則比卡超存在” is true.**

| p | q | $p \supset q$ |
|---|---|---------------|
| T | T | T             |
| T | F | F             |
| F | T | T             |
| F | F | T             |

# Conditional

Reasons:

1. Ensure that the truth-value of " $(\phi \supset \phi)$ " is always T whether  $\phi$  is T or F.

E.g 如果美斯是女人，則美斯是女人。

| p | q | $p \supset q$ |
|---|---|---------------|
| T | T | T             |
| T | F | F             |
| F | T | T             |
| F | F | T             |



# Conditional

Any cases that we endorse that a conditional is true if the antecedent is false ?

## Example given by Copi

Given  $2 < 4$ . From this, we know that all number which is small than 2 will be smaller than 4,

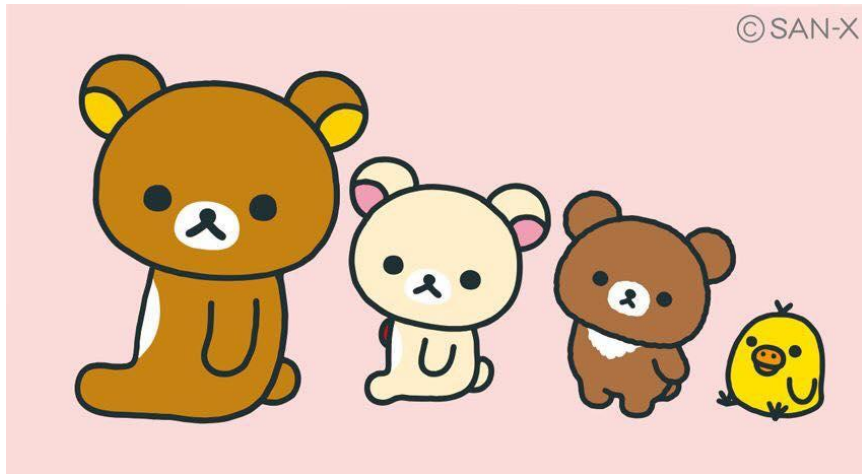
If  $x < 2$ , then  $x < 4$ .

Therefore,

If  $3 < 2$ , then  $3 < 4$ .

If  $5 < 2$ , then  $5 < 4$ .

# A Further Example



鬆弛熊高過蜜茶熊. 由此得知任何高過鬆弛熊的生物都比蜜茶熊高

If A 比鬆弛熊高, then A比蜜茶熊高.

所以,

If白熊比鬆弛熊高, then白熊比蜜茶熊高。

If小雞比鬆弛熊高, then小雞比蜜茶熊高

# Bi-conditional

To translate “if and only if”  
(iff)

June gets an A iff she gets full-  
mark. ( $P \equiv Q$ )

- June gets an A. ( $P$ )
- June gets full-mark. ( $Q$ )

| $p$ | $q$ | $p \equiv q$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | T            |

# Bi-conditional



It means  $(P \supset Q) \cdot (Q \supset P)$ .

The whole bi-conditional is true when and only when the antecedent and the consequent are of the **same truth value**.

# Well-formed Formulas(Wffs)

*Wff: Syntactically correct arrangement of symbols*

## Rules for Well-formed Formulas :

1. Capital letters are wffs.
2. If  $\alpha$  is a wff, then so is  $\sim\alpha$ .
3. If  $\alpha$  and  $\beta$  are wffs, then so is  $(\alpha\cdot\beta)$ .
4. If  $\alpha$  and  $\beta$  are wffs, then so is  $(\alpha\vee\beta)$ .
5. If  $\alpha$  and  $\beta$  are wffs, then so is  $(\alpha\supset\beta)$ .
6. If  $\alpha$  and  $\beta$  are wffs, then so is  $(\alpha\equiv\beta)$ .
7. Nothing else counts as a wff.

# Operators

Consider  $x \times y$ , suppose  $x = 1$  or  $0$  and  $y = 1$  or  $0$

| $x$ | $y$ | $x \times y$ |
|-----|-----|--------------|
| 1   | 1   | 1            |
| 1   | 0   | 0            |
| 0   | 1   | 0            |
| 0   | 0   | 0            |

| $p$ | $q$ | $p \cdot q$ |
|-----|-----|-------------|
| T   | T   | T           |
| T   | F   | F           |
| F   | T   | F           |
| F   | F   | F           |

you may also substitute **(a + b) into x**

# Operators

Consider  $(x + y) \times y$ ,

suppose  $x = 1$  or  $0$  and  $y = 1$  or  $0$

| x | y | $(x + y)$ | $\times$ | y |
|---|---|-----------|----------|---|
| 1 | 1 | 2         | 2        | 1 |
| 1 | 0 | 1         | 0        | 0 |
| 0 | 1 | 1         | 1        | 1 |
| 0 | 0 | 0         | 0        | 0 |

| P | Q | $(P \supset Q)$ | $\cdot$ | Q |
|---|---|-----------------|---------|---|
| T | T | T               | T       | T |
| T | F | F               | F       | F |
| F | T | T               | T       | T |
| F | F | T               | F       | F |

# Operators and Brackets

- In a compound statement, it may includes more than one logical operator.
  - ◆  $\sim(P \equiv (Q \vee R)) \supset (\sim R \cdot S)$
- Brackets are used to **avoid ambiguity**.
- Among those operators in the formulas, **one of them is the main operator**, and the rest are minor operators.



# Operators and Brackets

- The main operator is the operator that has everything else in the statement as its scope.
- The minor operators are the operators that have only the component of the whole statement as its scope.

$$\sim(P \equiv (Q \vee R)) \supset (\sim R \cdot S)$$

The diagram illustrates the scope of operators in the logical expression  $\sim(P \equiv (Q \vee R)) \supset (\sim R \cdot S)$ . Brackets are used to show the scope of each operator:

- A blue bracket under  $\vee$  indicates its scope is  $(Q \vee R)$ .
- A green bracket under  $\sim$  indicates its scope is  $R$ .
- A red bracket under  $\cdot$  indicates its scope is  $(\sim R \cdot S)$ .
- A purple bracket under  $\equiv$  indicates its scope is  $(P \equiv (Q \vee R))$ .
- An orange bracket under  $\supset$  indicates its scope is the entire expression  $\sim(P \equiv (Q \vee R)) \supset (\sim R \cdot S)$ , identifying it as the main operator.

# Operators and Brackets

□ An unary operator is an operator that operates on exactly one wff.

□ A binary operator is an operator that operates on exactly two wffs.

| Unary        | Binary                                                                                         |
|--------------|------------------------------------------------------------------------------------------------|
| $\sim$       | $\cdot, \vee, \supset, \equiv$                                                                 |
| $\sim\alpha$ | $\alpha \cdot \beta$<br>$\alpha \vee \beta$<br>$\alpha \supset \beta$<br>$\alpha \equiv \beta$ |

# Discussion 1

Which of the following formulas are wffs?

1.  $\sim\sim\sim P \vee R$

2.  $\sim PQ \vee R$

3.  $A \subset B$

4.  $A \sim B$

5.  $p \vee q$

6.  $P \supset \sim R$

7.  $Z$

8.  $\supset \sim R \cdot S$

9.  $M \equiv N \equiv Q$

10.  $((P \supset R) \cdot (S \vee R)) \equiv S$

11.  $P \sim \supset R$

12.  $M \leftrightarrow Q$

# Truth Table

Truth table: An arrangement of truth values that shows in every possible case how the truth value of a compound proposition is determined by the truth values of its simple components.

One row = one logical possibility

| P | Q | $P \supset Q$ |
|---|---|---------------|
| T | T | T             |
| T | F | F             |
| F | T | T             |
| F | F | T             |

# Truth Table

To construct a complete truth table,

1. Enter the truth values of the simple components.
2. Compute the truth values of the compound components.

**\*\* we have to start from the components in the innermost brackets.**

We have to determine the **number of rows** of it first .  
But how?

# Procedures

The number of rows =  $2^n$ , where  $n$  is the number of different simple proposition components.

The next step is to divide the number of lines in half. Then *we assign T to the half of the lines ( $2^n/2$  lines) for each letter and F on the remaining lines.*

Circle the column of main operator

**\*\*Make sure every logical combination is assigned to the truth table**

# Example 1

Construct a truth table for:

$$\sim(P \cdot (Q \equiv \sim P))$$

| P | Q | $\sim (P \cdot (Q \equiv \sim P))$ |
|---|---|------------------------------------|
|   |   |                                    |
|   |   |                                    |
|   |   |                                    |
|   |   |                                    |

# Example 1

Construct a truth table for:

$$\sim(P \cdot (Q \equiv \sim P))$$

| P | Q | $\sim$ | (P | . | (Q | $\equiv$ | $\sim$ | P)) |
|---|---|--------|----|---|----|----------|--------|-----|
| T | T | T      | T  | F | T  | F        | F      | T   |
| T | F | F      | T  | T | F  | T        | F      | T   |
| F | T | T      | F  | F | T  | T        | T      | F   |
| F | F | T      | F  | F | F  | F        | T      | F   |



# Discussion 1

Construct a truth table for:

$$\sim P \vee (\sim Q \supset P)$$

| P | Q | $\sim P \vee (\sim Q \supset P)$ |
|---|---|----------------------------------|
| T | T |                                  |
| T | F |                                  |
| F | T |                                  |
| F | F |                                  |

# Discussion 1

| P | Q | $\sim$ | P | $\vee$ | $(\sim$ | Q | $\supset$ | P) |
|---|---|--------|---|--------|---------|---|-----------|----|
| T | T | F      | T | T      | F       | T | T         | T  |
| T | F | F      | T | T      | T       | F | T         | T  |
| F | T | T      | F | T      | F       | T | T         | F  |
| F | F | T      | F | T      | T       | F | F         | F  |

**UGED1111 Logic**

**Symbolic Language  
and  
Truth Table II**

# Logical Relations and Properties of Propositions

## Logical Properties of propositions

1. Tautologous
2. Self-contradictory
3. Contingent

## Logical relations of propositions:

1. Logically Equivalent
2. Contradictory
3. Consistent
4. Inconsistent

# Properties of Propositions

## Tautologous

- **Necessarily true**
- **it is true regardless of the truth values of its components**
- **True in every possible interpretation , not just in actual world**
- **E.g.  $P \vee \sim P$**

# Properties of Propositions

## Self-contradictory

- **Necessarily false**
- **it is false regardless of the truth values of its components**
- **false in every logical possibility , not just in actual world**
- **E.g.  $P \cdot \sim P$**

# Properties of Propositions

## Contingent

- Its truth value varies depending on the truth values of its components.
- True in some logical possibilities; false in some logical possibilities
- i.e. at least one line true and at least one line false
- E.g.  $P \cdot Q$

# Discussion 3

**What is the logical property of the following propositions?**

1.  $(P \vee Q) \equiv \sim Q$
2.  $(P \supset Q) \cdot (\sim Q \cdot P)$



# Answer 3.1

| P | Q | $(P \vee Q) \equiv \sim Q$ |
|---|---|----------------------------|
| T | T | F                          |
| T | F | T                          |
| F | T | F                          |
| F | F | F                          |

**Contingent**

# Answer 3.2

| P | Q | (P $\supset$ Q) $\cdot$ ( $\sim$ Q $\cdot$ P) |   |   |   |   |   |   |   |
|---|---|-----------------------------------------------|---|---|---|---|---|---|---|
| T | T | T                                             | T | T | F | F | T | F | T |
| T | F | T                                             | F | F | F | T | F | T | T |
| F | T | F                                             | T | T | F | F | T | F | F |
| F | F | F                                             | T | F | F | T | F | F | F |

Self-contradictory

# Relations of Propositions

## Logically Equivalent

For any wffs  $\alpha$  and  $\beta$ ,  $\alpha$  and  $\beta$  are logically equivalent just in case they have the same truth value in every logical possibility.

| P | Q | $\sim$ P | $\vee$ Q | / | P | $\supset$ Q |
|---|---|----------|----------|---|---|-------------|
| T | T | F        | T        |   | T | T           |
| T | F | F        | F        |   | T | F           |
| F | T | T        | T        |   | F | T           |
| F | F | T        | T        |   | F | T           |

# Relations of Propositions

## Contradictory

For any wffs  $\alpha$  and  $\beta$ ,  $\alpha$  and  $\beta$  are contradictory just in case  $\alpha$  and  $\beta$  have different truth values in every logical possibility.

| P | Q | $\sim$ Q | $\cdot$ P | / P | $\supset$ Q |
|---|---|----------|-----------|-----|-------------|
| T | T | F        | F         | T   | T           |
| T | F | T        | T         | T   | F           |
| F | T | F        | F         | F   | T           |
| F | F | T        | F         | F   | T           |

# Relations of Propositions

## Consistent

For any wffs  $\alpha$  and  $\beta$ ,  $\alpha$  and  $\beta$  are consistent just in case there is at least one logical possibility that  $\alpha$  and  $\beta$  are both true.

| P | Q | $\sim$ Q | $\vee$ P | / | P | $\supset$ Q |
|---|---|----------|----------|---|---|-------------|
| T | T | F        | T        | T | T | T           |
| T | F | T        | T        | T | F | F           |
| F | T | F        | F        | F | T | T           |
| F | F | T        | T        | F | F | F           |

# Relations of Propositions

## Inconsistent

For any wffs  $\alpha$  and  $\beta$ ,  $\alpha$  and  $\beta$  are inconsistent just in case there is no logical possibility that  $\alpha$  and  $\beta$  are both true.

| P | Q | $\sim$ | Q | $\equiv$ | P | / | P | $\cdot$ | Q |
|---|---|--------|---|----------|---|---|---|---------|---|
| T | T | F      | T | F        | T | T | T | T       | T |
| T | F | T      | F | T        | T | T | F | F       | F |
| F | T | F      | T | T        | F | F | F | T       | T |
| F | F | T      | F | F        | F | F | F | F       | F |

# Discussion 4

**What is/are the logical relations of the following propositions?**

- 1.  $P \supset Q / \sim Q \cdot P$**
- 2.  $P \vee (Q \supset S) / S \vee (Q \supset P)$**

# Answer

| P | Q | (P $\supset$ Q) / ( $\sim$ Q $\cdot$ P) |   |   |   |   |   |   |
|---|---|-----------------------------------------|---|---|---|---|---|---|
| T | T | T                                       | T | T | F | T | F | T |
| T | F | T                                       | F | F | T | F | T | T |
| F | T | F                                       | T | T | F | T | F | F |
| F | F | F                                       | T | F | T | F | F | F |

**Contradictory and inconsistent**

**\*\*If two propositions are **Contradictory**, they must be **inconsistent**.**



# Answer

| P | Q | S | $P \vee (Q \supset S) / S \vee (Q \supset P)$ |
|---|---|---|-----------------------------------------------|
| T | T | T |                                               |
| T | T | F |                                               |
| T | F | T |                                               |
| T | F | F |                                               |
| F | T | T |                                               |
| F | T | F |                                               |
| F | F | T |                                               |
| F | F | F |                                               |

# Answer

| P | Q | S | P | $\vee$ | $(Q \supset S)$ | / | S | $\vee$ | $(Q \supset P)$ |
|---|---|---|---|--------|-----------------|---|---|--------|-----------------|
| T | T | T | T | T      | T               | T | T | T      | T               |
| T | T | F | T | T      | F               | F | F | T      | T               |
| T | F | T | T | T      | F               | T | T | T      | T               |
| T | F | F | T | T      | F               | F | F | T      | T               |
| F | T | T | F | T      | T               | T | T | T      | F               |
| F | T | F | F | F      | T               | F | F | F      | F               |
| F | F | T | F | T      | F               | T | T | T      | F               |
| F | F | F | F | T      | F               | T | F | T      | F               |

Logically Equivalent and consistent

# Validity

Checking the validity of an argument with truth table:

1. In a truth table for argument, we need to **list out all the possible truth values of premises and conclusion in the truth table.**
2. To clearly indicate premises and conclusion, we use “ / ” to separate the premises and “// ” to separate the last premise and the conclusion.
3. If you can find **any rows showing all true premises and false conclusion** , then the argument is invalid. (Indicate that row when you find it. )

# Discussion 4



Use a complete truth table to check whether the argument is valid.

1.  $P \supset Q / \sim P // Q$

# Answer

| P | Q | P $\supset$ Q | / | $\sim$ P | // | Q |
|---|---|---------------|---|----------|----|---|
| T | T | T             |   | F        |    | T |
| T | F | F             |   | F        |    | F |
| F | T | T             |   | T        |    | T |
| F | F | T             |   | T        |    | F |

Invalid

# Discussion 5

Use a complete truth table to check whether the argument is valid.

1.  $\sim T \supset S / \sim S \vee (\sim T \supset R) / \sim T // R$

# Answer

| R | S | T | $\sim$ | T | $\supset$ | S | / | $\sim$ | S | $\vee$ | ( $\sim$ | T | $\supset$ | R) | / | $\sim$ | T | // | R |
|---|---|---|--------|---|-----------|---|---|--------|---|--------|----------|---|-----------|----|---|--------|---|----|---|
| T | T | T | F      | T | T         | T |   | F      | T | T      | F        | T | T         | T  |   | F      | T |    | T |
| T | T | F | T      | F | T         | T |   | F      | T | T      | T        | F | T         | T  |   | T      | F |    | T |
| T | F | T | F      | T | T         | F |   | T      | F | T      | F        | T | T         | T  |   | F      | T |    | T |
| T | F | F | T      | F | F         | F |   | T      | F | T      | T        | F | T         | T  |   | T      | F |    | T |
| F | T | T | F      | T | T         | T |   | F      | T | T      | F        | T | T         | F  |   | F      | T |    | F |
| F | T | F | T      | F | T         | T |   | F      | T | F      | T        | F | F         | F  |   | T      | F |    | F |
| F | F | T | F      | T | T         | F |   | T      | F | T      | F        | T | T         | F  |   | F      | T |    | F |
| F | F | F | T      | F | F         | F |   | T      | F | T      | T        | F | F         | F  |   | T      | F |    | F |

Valid

# Case

$$(\mathbf{N} \cdot \mathbf{P} \equiv (\mathbf{Q} \vee \mathbf{R})) \supset (\sim \mathbf{S} \cdot \mathbf{M}) / (\mathbf{N} \cdot \mathbf{P}) / \mathbf{Q} // (\sim \mathbf{S} \cdot \mathbf{M})$$

Simple proposition as component: 6

We have to list  $2^6$  (i.e. 64 ) rows!

Indirect truth table saves our time!



# Indirect Truth Table

Indirect truth tables provide a faster way for checking validity of arguments.

◆ The idea is to check whether the argument is possible to have all true premises and a false conclusion by making assumptions.

◆ If it is possible for the argument to have all true premises and a false conclusion, then the argument is **invalid**. Otherwise, **valid**.

# Indirect Truth Table

We need to first **assume the conclusion is false and the premises are true**( which means it is invalid).

Then we have to see if there is any **contradiction** in the computation.

➤ If there is a contradiction, then the assumption is false , i.e. the argument is **valid**.

➤ If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is **invalid**.

# Example

## Whether the following argument is valid?

$$\sim(P \vee R)/Q \equiv \sim P//Q \cdot \sim R$$

[illegible]

# Case 1 : $\sim(P \vee R)/Q \equiv \sim P//Q \cdot \sim R$

- If there is a contradiction, then the assumption is false , i.e. the argument is **valid**.
- If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is **invalid**.
  - In the indirect truth table of this argument ,there is a contradiction.
  - Indicate the contradiction by circling it.
  - It is **valid**!!!!

| $\sim$ | (P | $\vee$ | R) | / | Q | $\equiv$ | $\sim$ | P | // | Q | $\cdot$ | $\sim$ | R |
|--------|----|--------|----|---|---|----------|--------|---|----|---|---------|--------|---|
| T      | F  | F      | F  |   | F | T        | F      | T |    | F | F       | T      | F |
| ①      | ③  | ②      | ③  |   | ⑦ | ①        | ⑧      | ⑨ |    | ⑥ | ①       | ⑤      | ④ |

# Indirect Truth Table

| $(P \equiv Q) \supset S$ | $/$ | $\sim S$ | $\cdot$ | $P$ | $//$ | $P \vee Q$ |
|--------------------------|-----|----------|---------|-----|------|------------|
| F                        | T   | F        | T       | T   | F    | F          |

Similar techniques you may  
have used before

SUDOKU!!!!!!

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 4 | 8 |   |   | 6 | 1 |   |   | 2 |
|   |   | 2 | 3 |   |   |   |   | 1 |
| 1 | 9 |   |   |   | 7 | 6 | 4 | 5 |
|   | 1 | 6 |   | 2 |   | 5 |   | 9 |
| 3 |   | 7 | 1 |   | 8 | 4 |   | 6 |
| 9 |   | 4 |   | 7 |   | 3 | 1 |   |
| 2 | 3 | 1 | 8 |   |   |   | 5 | 7 |
| 7 |   |   |   |   | 2 | 1 |   |   |
| 5 |   |   | 7 | 1 |   |   | 6 | 3 |

# Indirect Truth Table

□ Sometimes we need to make further assumption if the truth value of certain letters are not fixed by previous steps.

| P | $\supset$ | Q | / | Q | $\supset$ | P | // | P | $\equiv$ | Q |
|---|-----------|---|---|---|-----------|---|----|---|----------|---|
| T | T         | F |   | F | T         | T |    | T | F        | F |
| 3 | 1         | 4 |   | 4 | 1         | 3 |    | 2 | 1        | 2 |
| F | T         | T |   | T | T         | F |    | F | F        | T |
| 3 | 1         | 4 |   | 4 | 1         | 3 |    | 2 | 1        | 2 |

□ If no contradiction is found, then the argument is invalid, you can stop further computation.

□ If a contradiction has been found, you **CANNOT** stop the computation. **You need to exhaust every logical possibility.**

# Reference

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- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.

**UGED1111 Logic**

**Natural Deduction I**

CHAN, Hei Man



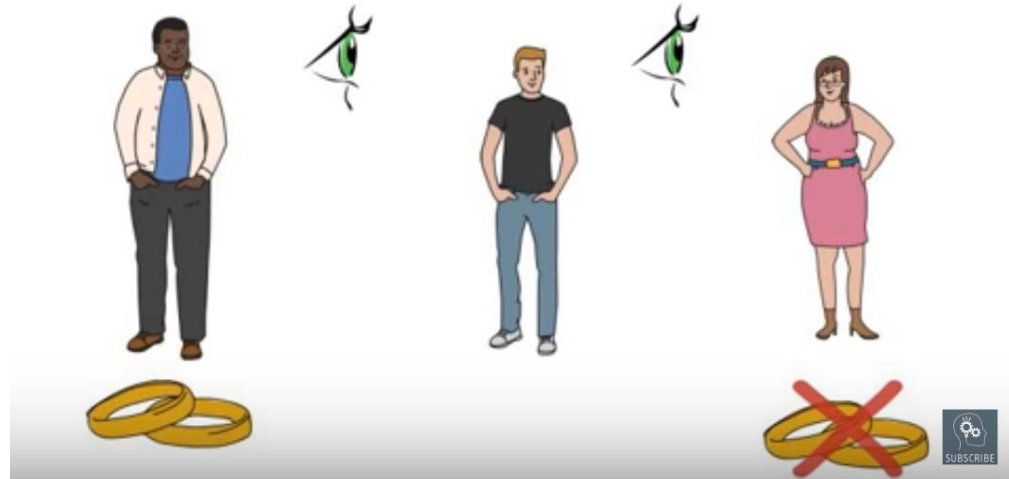
# Scenario

Is a married person looking at an unmarried person?

A: Yes

B: No

C: Not enough  
information



# Scenario

1. 當小蘭在廚房或小東不在廚房，則廚師會在廚房。
  2. 只要小東在廚房，則小北也在廚房。。
  3. 並不是小北或小七在廚房。
  4. 廚師在廚房。
- 

1.  $(P \vee \sim R) \supset S$
2.  $R \supset Q$
3.  $\sim (Q \vee M) // S$

P:小蘭在廚房。  
Q:小北在廚房。  
R:小東在廚房。  
S:廚師在廚房。  
M: 小七在廚房



# Checking validity

| (P | v | $\sim R) \supset S / R \supset Q / \sim (Q \vee M) // S$ |
|----|---|----------------------------------------------------------|
|    |   |                                                          |

Valid argument.

But is this all that we want to know???

We also want to know how he deduces the conclusion from the premises!

# Natural Deduction



- ❑ Natural deduction is a method of deriving the conclusion of a valid argument from the premises through a series of discrete steps.
- ❑ This method aims to demonstrate the reasoning process of an argument (how we prove the conclusion from the premises).
- ❑ However, it does not work for invalid arguments.

# 18 Rules of Inference

**There are 18 rules of inference.**

□ **8 Rules of Implication**

**+**

□ **10 Rules of Replacement**

**\*\*\*All 18 rules will be given in the exam. So you don't need to bother memorizing them.**

# Rules of implication I

1. Modus Ponens (MP)

$p \supset q$

$p$

---

$q$

2. Modus Tollens (MT)

$p \supset q$

$\sim q$

---

$\sim p$

3. Hypothetical Syllogism (HS)

$p \supset q$

$q \supset s$

---

$p \supset s$

4. Disjunctive Syllogism (DS)

$p \vee q$

$\sim p$

---

$q$

# Rules of implication II

5. Constructive Dilemma (CD)

$(p \supset q) \cdot (r \supset s)$

$p \vee r$

---

$q \vee s$

6. Simplification (Simp)

$p \cdot q$

---

$p$

7. Conjunction (Conj)

$p$

$q$

---

$p \cdot q$

8. Addition (Add)

$p$

---

$p \vee q$

# 1. Modus Ponens (MP)

Given a conditional statement , we can deduce its consequent by asserting its antecedent.

$p \supset q$

$p$

---

$q$

1. If you get GPA 4 , you will get the scholarship ◦

2. You get GPA 4 ◦

---

3. You will get the scholarship ◦



## 2. Modus Tollens (MT)

Given a conditional statement and the negation of its consequent, then we can deduce the negation of its antecedent.

$$\begin{array}{l} p \supset q \\ \sim q \\ \hline \sim p \end{array}$$

1. If the temperature is higher than 30 degree Celsius , then the school will turn on the air-conditioner .
2. The school did not turn on the air-conditioner today .

---
3. The temperature is not higher than 30 degree Celsius today.

### 3. Hypothetical Syllogism (HS)

Given two conditional statements with a common formula as transition then we can deduce a new conditional without that common formula.

$$p \supset q$$
$$q \supset s$$

---

$$p \supset s$$

1. 如果我收到利息，則母親幫我保存利息。
  2. 如果母親幫我保存利息，則利息變成母親存款。
- 
3. 如果我收到利息，則利息變成母親存款。

# 4. Disjunctive Syllogism (DS)

Given a disjunctive statement and the negation of its left-hand disjunct, we can deduce the right-handed disjunct.

$$p \vee q$$
$$\sim p$$

---

$$q$$

1. 小明是帥哥或者有才華的。

2. 小明不是帥哥。

---

3. 小明是有才華的。

# Discussion 1a

Construct a formal proof of validity for the following arguments by Natural Deduction.

1.  $P \supset Q$
2.  $Q \supset S$
3.  $\sim S \quad // \sim P$

# Discussion 1b

Construct a formal proof of validity for the following arguments by Natural Deduction.

1.  $P \vee S$
2.  $S \supset Q$
3.  $\sim P \quad //Q$

# 5. Constructive Dilemma (CD)

Given a conjunction of two conditional statements.  
If either of the antecedents is true, then either of  
the consequents is true.

$(p \supset q) \cdot (r \supset s)$

$p \vee r$

---

$q \vee s$

1. 如果下雨則賣水鞋，並且如果天晴則賣傘。

2. 下雨或天晴。

---

3. 賣水鞋或賣傘。

# 6. Simplification (Simp)

Given a conjunction of two statement, then we can deduce the left-sided conjunct.

$$\frac{p \cdot q}{p}$$

1. 殺老師反應很快，而且他很聰明。

2. 殺老師反應很快。



# 7. Conjunction (Conj)

Given that two propositions are true separately,  
then their conjunction are also true.

$p$

1. I have a pen.

$q$

2. I have an apple.

---

$p \cdot q$

---

3. I have a pen and a apple.

AR!Apple pen!





# 8. Addition (Add)

Given a proposition is true, its disjunction with any proposition is also true.

$p$

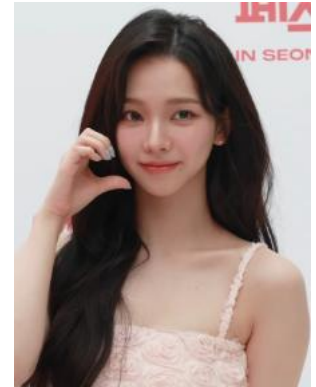
---

$p \vee q$

1. Karina很漂亮。

---

2. Karina很漂亮，或者Karina很可愛。



# Discussion 2

Construct a formal proof of validity for the following arguments by Natural Deduction.

1.  $P \cdot M$
2.  $P \supset Q$
3.  $R \supset S // Q \vee S$

# Exercise 1

1.  $\sim C \supset (A \supset C)$

2.  $\sim C$   $// \sim A$

# Exercise 2

1.  $\sim S \supset D$

2.  $\sim S \vee (\sim D \supset K)$

3.  $\sim D$  // K

# Invalid inference

**Rules of implication cannot be applied on a part of compound statements.**

1.  $P \supset Q$   
2.  $(P \vee R) \supset Q$  1, Add

1.  $(P \cdot R) \supset Q$   
2.  $P \supset Q$  1, Simp

1.  $P \supset Q$   
2.  $(P \supset Q) \vee R$  1, Add

1.  $(P \supset Q) \cdot R$   
2.  $P \supset Q$  1, Simp

# Improper use of rules

**Even though the following inferences are valid, they are improper.**

1.  $P$   
2.  $Q \vee P$       1, Add

1.  $P \cdot R$   
2.  $R$       1, Simp

1.  $P$   
2.  $P \vee Q$       1, Add

1.  $P \cdot R$   
2.  $P$       1, Simp

# Rules of replacement I

9. DeMorgan's Rule (DM)

$$\sim(p \cdot q) :: \sim p \vee \sim q$$

$$\sim(p \vee q) :: \sim p \cdot \sim q$$

10. Commutativity (Com)

$$p \vee q :: q \vee p$$

$$p \cdot q :: q \cdot p$$

11. Association (Assoc)

$$p \vee (q \vee r) :: (p \vee q) \vee r$$

$$p \cdot (q \cdot s) :: (p \cdot q) \cdot s$$

12. Distribution (Dist)

$$p \cdot (q \vee r) :: (p \cdot q) \vee (p \cdot r)$$

$$p \vee (q \cdot s) :: (p \vee q) \cdot (p \vee s)$$

13. Double Negation (DN)

$$p :: \sim\sim p$$

# 9. DeMorgan's Rule (DM)

**“Not both p and q” is logically equivalent to “Not p or not q” ;  
and**

**“Not either p or q” is logically equivalent to “Not p and not q.”**

$$\sim (p \cdot q) :: \sim p \vee \sim q$$

並不是大嘅鬼和「任何人」都有頭髮。  
大嘅鬼沒有頭髮或「任何人」沒有頭髮。

$$\sim (p \vee q) :: \sim p \cdot \sim q$$

並不是大嘅鬼或「任何人」有穿衣服。  
大嘅鬼沒有穿衣服而且「任何人」沒有穿衣服。





# 10. Commutativity (Com):

The truth value of a conjunction or disjunction is unaffected by the order in which the components are listed.

$$p \vee q :: q \vee p$$

今天是晴天，或者今天是陰天。  
今天是陰天，或者今天是晴天。

$$p \cdot q :: q \cdot p$$

熊本熊和鬆弛熊來中大。  
鬆弛熊和熊本熊來中大。

# 11. Association (Assoc)

When the same operator is used through out a conjunctive or disjunctive statement, the placement of brackets will not alter the truth value of the whole statement.

$$p \vee (q \vee r) :: (p \vee q) \vee r$$

今天是晴天，或者今天是陰天或雨天。  
今天是晴天或陰天，或者今天是雨天。

$$p \cdot (q \cdot s) :: (p \cdot q) \cdot s$$

北極熊來中大，而且黑熊和啡熊來中大。  
北極熊和黑熊來中大，而且啡熊來中大。

# 12. Distribution (Dist):

When a proposition is conjoined to a disjunctive statement in bracket or disjoined to a conjunctive statement in bracket, the rule allows us to put that proposition together with each of the components inside the bracket, and also to go in the reverse direction.

$$p \cdot (q \vee r) :: (p \cdot q) \vee (p \cdot r)$$

北極熊來中大，並且黑熊或啡熊來中大。  
北極熊和黑熊來中大，或者北極熊和啡熊來中大。

$$p \vee (q \cdot s) :: (p \vee q) \cdot (p \vee s)$$

**Similar technique you have used before**

$$2 \times (4+3) = (2 \times 4) + (2 \times 3)$$

$$m \times (n+y) = mn + my$$

# 13. Double Negation (DN)

If pairs of tildes immediately adjacent to one another, deleting or introducing them cannot change the truth value of the whole statement.

$p :: \sim\sim p$

我沒有不上課。  
我有上課。

負負得正?!

# Discussion 3



1.  $\sim (\sim P \vee Q)$

2.  $P \supset S \quad //S$

# Discussion 4



1.  $D \bullet (E \vee F)$

2.  $\sim D \vee \sim F \quad // \quad D \bullet E$

# Exercise 3

1.  $(K \cdot H) \vee (K \cdot L)$

2.  $\sim L$   $// H$

# Exercise 4

1.  $S \vee (P \cdot Q)$
2.  $(S \vee Q) \supset \sim R$
3.  $(R \vee P) \vee Q \quad // P \vee Q$



# 14. Transposition (Trans):

The antecedent and consequent of a conditional statement may switch places if and only if tildes are inserted before both or tildes are removed from both.

$$(p \supset q) :: (\sim q \supset \sim p)$$

如果昨天我睡得夠，則我今天上課很清醒。  
如果我今天上課不是很清醒，則我昨天睡不夠。

# 15. Material Implication (Impl)

**Material implication** indicates the truth condition of a conditional statement, which means that a conditional statement is true if and only if the antecedent is false or the consequent is true

$$(p \supset q) :: (\sim p \vee q)$$

如果我口渴，則我喝水。  
我不是口渴，或者我喝水。

# 16. Exportation (Exp)

“If we have both p and q, then we have r”

is logically equivalent to

“If we have p, then if we have q, then we have r.”

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

如果你有車和有樓時，我們結婚。

如果你有車，則當你有樓時，我們結婚，。

# Recall

## Rule of replacement

如果你有車和有樓時，則我們結婚。

## Exportation

如果你有車，則當你有樓時，我們結婚，。

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

**Consider the following case of using rule of implication**

1.  $(P \cdot Q) \supset R$

2.  $P \supset R$                       1, Simp



1. 如果你有車和有樓時，我們結婚。

2. 如果你有車，我們結婚。

1, Simp

# 17. Material Equivalent (Equiv):

- (i) The first is the same as the definition of material equivalence given in the previous class.
- (ii) The second formulation indicates the truth condition of “ $p \equiv q$ ”, which means “ $p \equiv q$ ” is *true iff either  $p$  and  $q$  are both true or  $p$  and  $q$  are both false*”.

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

# 17. Material Equivalent (Equiv):

我獲得獎學金當且僅當我GPA 4.

(1)我獲得獎學金當我GPA 4，而且當我獲得獎學金則我GPA 4

(2)我獲得獎學金並且我GPA 4，或者我沒有獲得獎學金並且我不是GPA 4.

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

# 18. Tautology (Taut):

To eliminate redundancy in disjunctions and conjunctions.

$$p :: (p \vee p)$$

我好想睡或者我好想睡。

$$p :: (p \cdot p)$$

我好肚餓而且我好肚餓。

# Rules of replacement II

14. Transposition (**Trans**)

$$(p \supset q) :: (\sim q \supset \sim p)$$

15. Material Implication (**Impl**)

$$(p \supset q) :: (\sim p \vee q)$$

16. Exportation (**Exp**)

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

17. Material Equivalent (**Equiv**)

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

18. Tautology (**Taut**)

$$p :: (p \vee p)$$

$$p :: (p \cdot p)$$



# Discussion 5



1.  $S \supset (\sim S \vee P) \quad // \quad S \supset P$

# Discussion 6



1.  $P \equiv Q$
2.  $Q \supset (R \cdot S)$
3.  $\sim S \quad // \sim P$

# Discussion 7



1.  $P \supset Q$
2.  $S \supset M$
3.  $\sim Q \quad // \sim P \vee M$

# Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- <https://www.youtube.com/watch?v=llnyN-WD9u4>

# UGED1111 Logic

## Natural Deduction – Conditional Proof and Indirect Proof

# Translation

Simple statement/ proposition: P, Q, R, .....

Logical Operators :  $\sim$  ,  $\cdot$  ,  $\vee$  ,  $\supset$  ,  $\equiv$

**Example: All cats are cute.**

**Categorical Syllogism: All cats are cute animals.**

**Propositional Logic: P**

# Translation

1. Snow is white. (**P**)
2. The Show is not white. ( $\sim \mathbf{P}$ )
3. Both *Ironman* and *Ant-man* are Marvel Characters. ( $A \cdot B$ )
  - *Ironman* is a Marvel Character. ( $A$ )
  - *Ant-man* is a Marvel Character. ( $B$ )
4. Winnie are clever or beautiful. ( $M \vee N$ )
  - Winnie are clever. ( $M$ )
  - Winnie are Beautiful. ( $N$ )
5. If Joe feels bored, he read books about logic. ( $Y \supset W$ )
6. Joe feels bored iff he read books about logic. ( $Y \equiv W$ )
  - Joe feels bored. ( $Y$ )
  - Joe read books about logic. ( $W$ )

# Properties of Proposition

## Tautologous

- **Necessarily true**
- **e.g.  $P \vee \sim P$**
- e.g. Mimi is a cat or Mimi is not a cat.

## Contingent

e.g. Mimi is a cat

## Self-contradictory

- **Necessarily false**
- **e.g.  $P \cdot \sim P$**
- e.g. Mimi is a cat and Mimi is not a cat.



# Example

1.  $N \supset O$

2.  $N \supset P$

//  $N \supset (O \cdot P)$

# Conditional Proof

**Conditional proof** is a method for deriving a **conditional statement** (either the conclusion or some intermediate line) that offers a shorter and simpler way to derive the conclusion (comparing to direct method).

**\*\*Also**, some arguments have conclusions that **cannot** be derived by the direct method.  
That's why we need it.

# Conditional Proof



1. Assume the antecedent of the required conditional statement on one line (ACP step)
2. Derive the consequent on a subsequent line
3. “Discharge” this sequence of lines in a conditional statement that exactly replicates the targeted statement

**\*\*You can make more than one assumption**

# Example

1.  $N \supset O$

2.  $N \supset P$

//  $N \supset (O \bullet P)$

|

# Conditional Proof



Why can we make an assumption in the proof?

Do we assert a statement in doing the conditional proof?

NO!

Not asserting an statement as long as we discharge the assumption.

# Discussion

$$1. A \supset (B \bullet C)$$

$$2. (B \vee D) \supset E \quad // A \supset E$$

|

# Discussion

1.  $(G \vee H) \supset (S \bullet T)$

2.  $(T \vee U) \supset (C \bullet D) \quad // \quad G \supset C$

|

# Example

1.  $C \supset (A \bullet D)$

2.  $B \supset (A \bullet E)$

//  $(C \vee B) \supset A$

|

|

|



# Indirect Proof

Indirect proof can be used on any argument to derive either the conclusion or some intermediate line leading to the conclusion.

1. Assume the negation of the statement to be obtained.
2. Derive a contradiction from the assumption
3. Conclude that the original assumption is false
4. i.e. the targeted statement is true

\*\* Sometimes you may need to use DN to derive the targeted statement

# Example

1.  $(S \vee T) \supset \sim S$

//  $\sim S$

|

# Example

1.  $\sim M \supset (N \bullet O)$

2.  $N \supset P$

3.  $O \supset \sim P$                       // M

|

# Example

1. K

//  $S \supset S$

|

# Remarks



After discharging the assumption, you cannot use any formula derived from that assumption to further derive.

# Exercise

1.  $A \supset \sim B$

2.  $\sim(C \cdot \sim A)$                        $// C \supset \sim B$

# Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.

# Checking Validity of an argument with an indirect truth table



# Indirect Truth Table

Indirect truth tables provide a faster way for checking validity of arguments.

- ◆ The idea is to check whether the argument is possible to have all true premises and a false conclusion by making assumptions.
- ◆ If it is possible for the argument to have all true premises and a false conclusion, then the argument is **invalid**. Otherwise, **valid**.

# Indirect Truth Table

We need to first **assume the conclusion is false and all premises are true**( which means it is invalid).

Then we have to see if there is any **contradiction** in the computation.

➤ If there is a contradiction, then the assumption is false ,  
i.e. the argument is **valid**.

➤ If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is **invalid**.

# Case 1 : $\sim(P \vee R)/Q \equiv \sim P//Q \cdot \sim R$

- If there is a contradiction, then the assumption is false , i.e. the argument is **valid**.
- If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is **invalid**.
- In the indirect truth table of this argument ,there is a contradiction.
- **We need to indicate the contradiction by circling it.**
- It is **valid**!!!!

| $\sim$ | (P | $\vee$ | R) | / | Q | $\equiv$ | $\sim$ | P | // | Q | $\cdot$ | $\sim$ | R |
|--------|----|--------|----|---|---|----------|--------|---|----|---|---------|--------|---|
| T      | F  | F      | F  |   | F | T        | F      | T |    | F | F       | T      | F |
| ①      | ③  | ②      | ③  |   | ⑦ | ①        | ⑧      | ⑨ |    | ⑥ | ①       | ⑤      | ④ |

## Case 2: $P \supset S / (S \cdot \sim Q) \vee P // P \vee Q$

- We find no contradiction.
- It is an **invalid** argument since we can find a row (a logical possibility) with all true premises and a false conclusion.

| P | $\supset$ | S | / | (S | $\cdot$ | $\sim$ | Q) | $\vee$ | P | // | P | $\vee$ | Q |
|---|-----------|---|---|----|---------|--------|----|--------|---|----|---|--------|---|
| F | T         | T |   | T  | T       | T      | F  | T      | F |    | F | F      | F |
| 3 | 1         | 7 |   | 6  | 5       | 6      | 4  | 1      | 3 |    | 2 | 1      | 2 |

Exercise :  $A \cdot E / (A \vee D) \equiv \sim B // D \supset B$

Invalid

| $A \cdot E / (A \vee D) \equiv \sim B // D \supset B$ |   |   |  |   |   |   |   |   |   |  |   |   |   |
|-------------------------------------------------------|---|---|--|---|---|---|---|---|---|--|---|---|---|
| T                                                     | T | T |  | T | T | T | T | T | F |  | T | F | F |
| 3                                                     | 1 | 3 |  | 7 | 6 | 8 | 1 | 5 | 4 |  | 2 | 1 | 2 |

# Indirect Truth Table

□ Sometimes we need to make further assumption if the truth value of certain letters are not fixed by previous steps.

| P | $\supset$ | Q | / | Q | $\supset$ | P | // | P | $\equiv$ | Q |
|---|-----------|---|---|---|-----------|---|----|---|----------|---|
| T | T         | F |   | F | T         | T |    | T | F        | F |
| F | T         | T |   | T | T         | F |    | F | F        | T |

□ If no contradiction is found, then the argument is invalid, you can stop further computation.

□ If a contradiction has been found, you **CANNOT** stop the computation. **You need to exhaust every logical possibility.**

### Case 3: $P \supset Q / Q \supset P / P \equiv Q$

- Again assume that all premises are true and the conclusion is false.
- However, we cannot determine the truth value of “P” and “Q” without further assumptions
- “ $P \supset Q$ ” is true iff
  - (a) both “P” and “Q” are true ,or
  - (b) “P” is false and “Q” is true, or
  - (c) “P” is false and “Q” is false.
- For “ $Q \supset P$ ”, **there are three possibilities for it to be true.**
- For “ $P \equiv Q$ ”, **there are two possibilities for it to be false.**
- Therefore, we have to make further assumptions.

### Case 3: $P \supset Q / Q \supset P / P \equiv Q$

- We will start from “ $P \equiv Q$ ” since there are only 2 cases which “ $P \equiv Q$ ” is false instead of 3 cases to be true for the premises.
- “ $P \equiv Q$ ” is false iff
- (a) “ $P$ ” is true and “ $Q$ ” is false; or (b) “ $P$ ” is false and “ $Q$ ” is true;
- We will **assume “ $P$ ” is true and “ $Q$ ” is false first.**
- **If there is no contradiction, we can stop since we already found a line with all true premises and a false conclusion.**
- **But if there is a contradiction, we need to continue to look at other cases until we exhaust all the possibilities.**

| P $\supset$ Q / |  |  | Q $\supset$ P // |  |  | P $\equiv$ Q |   |   |
|-----------------|--|--|------------------|--|--|--------------|---|---|
| T               |  |  | T                |  |  | T            | F | F |
| 1               |  |  | 1                |  |  | 2            | 1 | 2 |
|                 |  |  |                  |  |  | F            | F | T |



### Case 3: $P \supset Q / Q \supset P // P \equiv Q$

- Since there is a contradiction in all cases, i.e. we cannot find a row (logical possibility) that all premises are true and the conclusion is false.
- Therefore, the argument is **valid**

| P | $\supset$ | Q | / | Q | $\supset$ | P | // | P | $\equiv$ | Q |
|---|-----------|---|---|---|-----------|---|----|---|----------|---|
| T | T         | F |   | F | T         | T |    | T | F        | F |
| 3 | 1         | 4 |   | 4 | 1         | 3 |    | 2 | 1        | 2 |
| F | T         | T |   | T | T         | F |    | F | F        | T |
| 3 | 1         | 4 |   | 4 | 1         | 3 |    | 2 | 1        | 2 |

# Example

Whether the following argument is valid?

$(P \equiv Q) \supset S$  /  $\sim S \cdot P$  /  $\therefore P \vee Q$

Valid

| $(P$ | $\equiv$ | $Q)$ | $\supset$ | $S$ | / | $\sim$ | $S$ | $\cdot$ | $P$ | // | $P$ | $\vee$ | $Q$ |
|------|----------|------|-----------|-----|---|--------|-----|---------|-----|----|-----|--------|-----|
| F    | T        | F    | T         | F   |   | T      | F   | T       | T   |    | F   | F      | F   |
| 5    | 8        | 7    | 1         | 6   |   | 2      | 4   | 1       | 2   |    | 3   | 1      | 3   |